Nowcasting

Instructions

Workshop 1 (Part 2) – Introduction to EViews Programming



IMF Institute for Capacity Development

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Section 1: EViews Fundamentals

EViews is an easy-to-use statistical, econometric, and economic modeling package, particularly suited for analyzing time-series objects (e.g., matrices).

There are three ways to work in EViews:

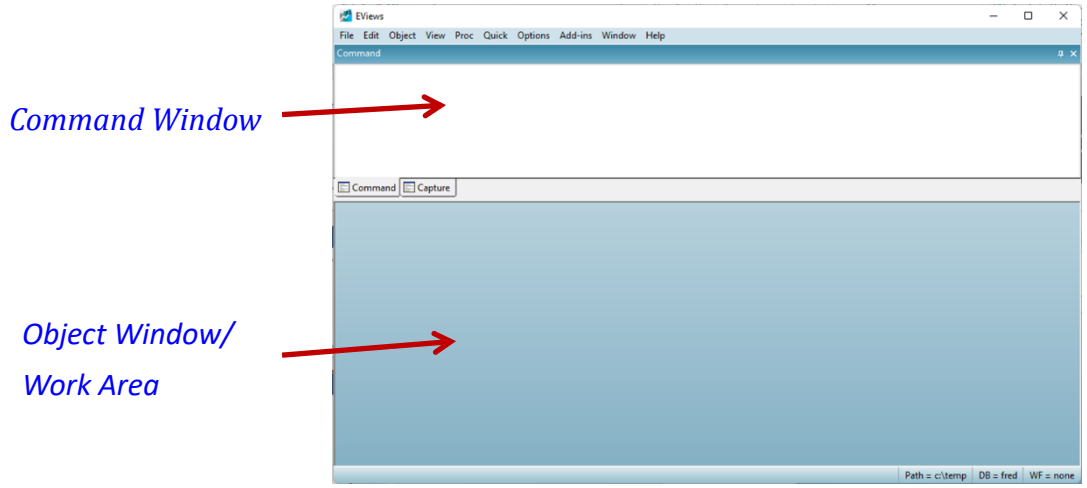
1) Using the graphical user interface (using mouse and menus/dialogs).

2) Issuing single commands (using the command window).

3) Executing program files (commands assembled in a script that is executed in batch mode).

1. STARTING EVIEWS

Double click on the "EViews" icon on your desktop. You should see the following window:



EViews is ready to accept typed commands, either in the "command line" window, which is immediately below the main menu, or by clicking on specific menu items on the main menu (e.g., "File") using the mouse.

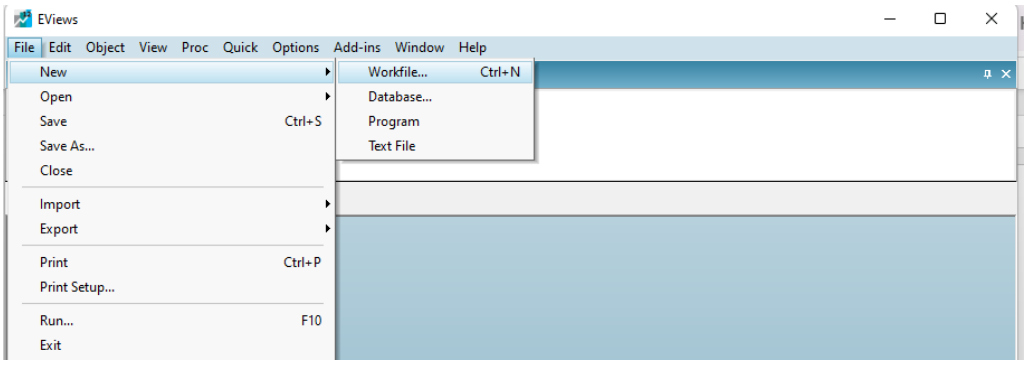
Any output (e.g., a plot, or regression output) arising from commands typed in the command window (e.g., plot) or by clicking on specific menu items will appear in the "Object Window/Work Area", positioned immediately below the command window.

2. CREATING AN EVIEWS WORKFILE

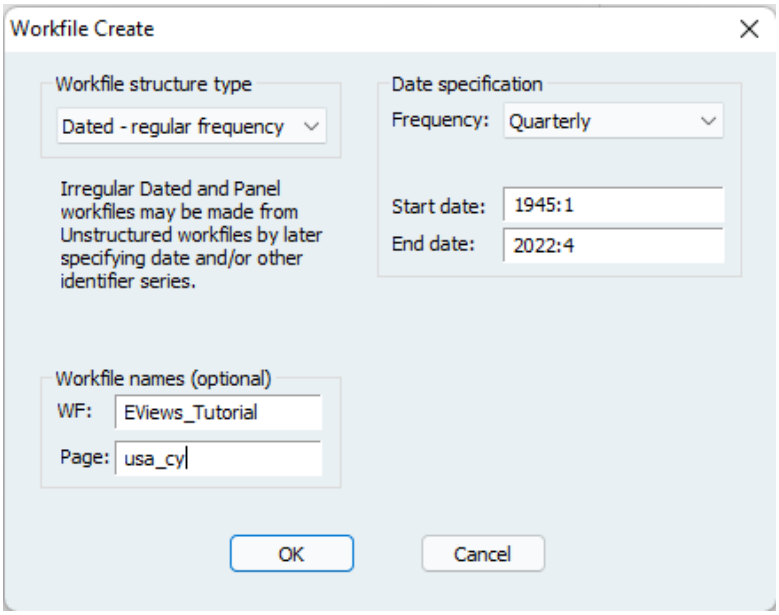
Before any commands or menu items can be selected/executed, one needs to create an EViews workfile. A workfile is a storage facility for the objects (e.g., a time-series, a regression, a plot) that you create during an EViews session. Workfiles can be created from either the "File" menu, or by issuing a specific EViews command. To use the file menu, invoke:

File → New → Workfile

namely:



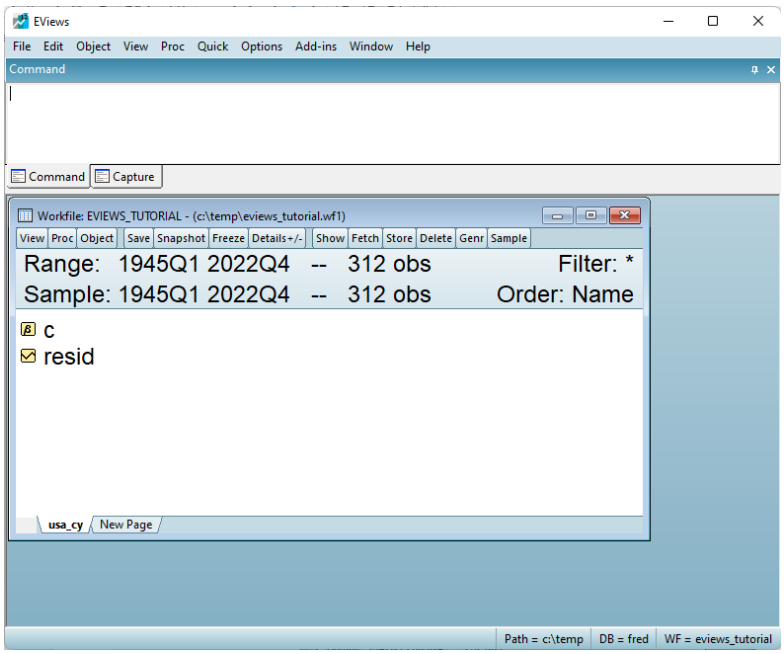
The dialog box that appears is shown on the next page. It allows you to specify the name of the workfile on your system, and the frequency of the first (or initial) pagefile in the workfile. An EViews workfile can contain multiple pagefiles, each with its own frequency.



In this case EViews is being directed to create a workfile called "**EViews_Tutorial**" together with the pagefile called "**usa_cy**" in which quarterly series will be saved. The earliest observation allowed is the first quarter of 1945 ("1945:1"). The latest is the fourth (or last quarter) of 2022 ("2022:4").

Note that a "pagefile" is simply a permanent collection of series and statistical objects (e.g., ordinary least squares) that are typically functions of the data saved in the pagefile.

Press "OK" to create the workfile. EViews should respond with the following view:



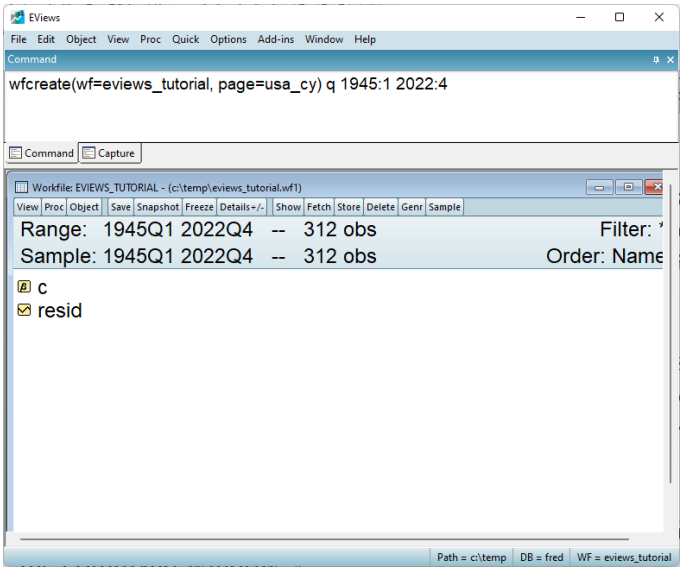
Notice that the newly created pagefile "**usa_cy**" already has two series associated with it: "**c**", containing the values of the most recent parameters estimated, and "**resid**", which will contain the estimated residuals of the most recent regression procedure you use. You will find these two series very useful for evaluating the regression results.

Note that double-clicking on the bottom tab labelled **"Path = "** allows you to change the default folder for saving workfiles. Double-clicking on the "WF" tab allows you to change the location (i.e., save it to a folder other than the default folder) and name of the workfile.

Lastly, it is possible to create the same workfile by issuing the following EViews command in the command window:

wfcreate(wf=eviews_tutorial, page=usa_cy) q 1945:1 2022:4

This command is especially useful when you wish to script your work in a program file that can be run on demand, making replication of your work far easier. EViews program files are discussed at the end of this guide.



3. CREATING AN EMPTY SERIES

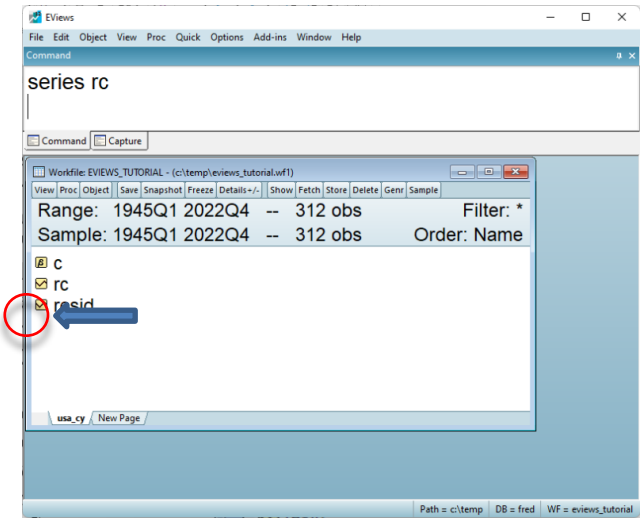
Easiest way to create an empty series is to type (in the command area) "**Series**" followed by the name[1] of the series you want to create.

For example, the command

series rc

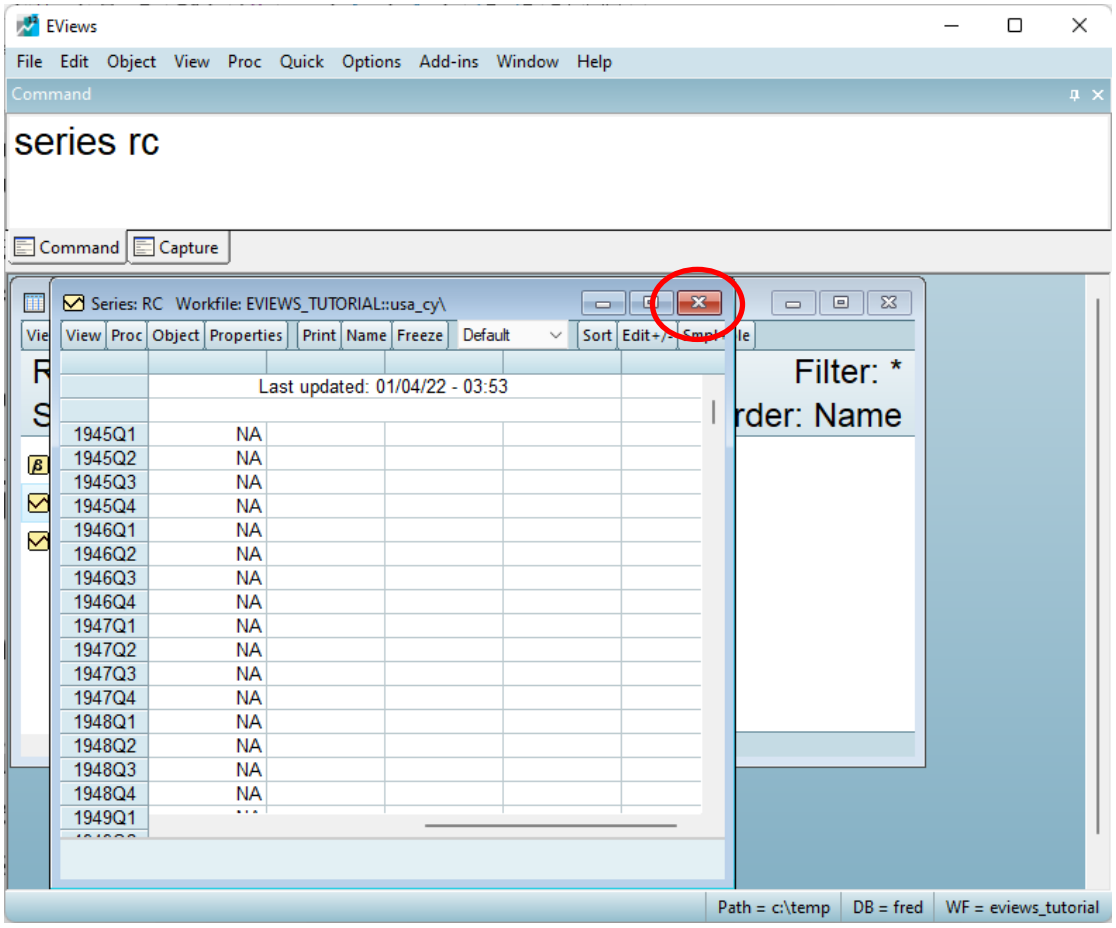
creates a time-series called "**rc**" in the active pagefile. It will have "NA" (EViews missing value code) for every value in the series from 1945:1 to 2022:4.

Notice that the workfile now has a new series called "**rc**" in the object names list.



[1] Series/object names can contain up to 24 characters but cannot have embedded white space.

Double-clicking on **rc** reveals the content of the series, viz:



A strong feature of EViews is its ability to handle missing values seamlessly in statistical operations.

You may close the spreadsheet view by clicking on the red button marked with an X: 

4. SAVING A WORKFILE

From the work/page file view, click on the "**Save**" button (to the right of "**View", "Proc", "Object"**)[2] to make your preferences/changes permanent (i.e., saved to the hard drive). The resulting file, which in this case is called **"Eviews_Tutorial"** will have an extension "**.wf1**".

Be sure to select "double-precision" for the data storage mode. This will help to minimize rounding errors.

You can also initiate a workfile save by clicking on the X button in the upper right-corner of the EViews screen. EViews will ask whether it should discard any unsaved workfiles.

[2] See workfile image at the top of page 6.

5. CREATING A WORKFILE FROM A FOREIGN DATA SOURCE (E.G., EXCEL)

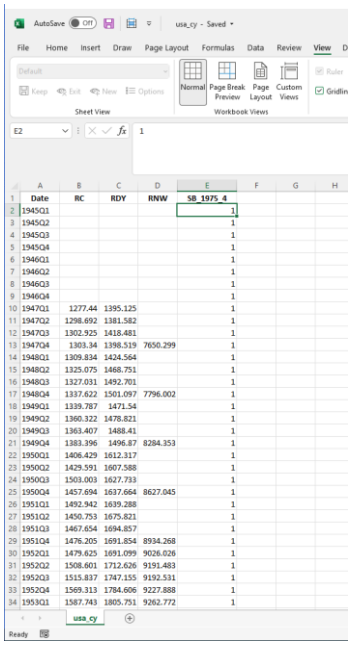
EViews can create a new workfile by opening data from a variety of formats (Excel ®, HTML, CSV, ASCII, RATS, Stata, SPSS, SAS, etc.). EViews automatically recognizes the format and structure of files when you open them. Simply open the file in EViews and let the application do the rest.
There are several ways to open a foreign file in EViews[3]:

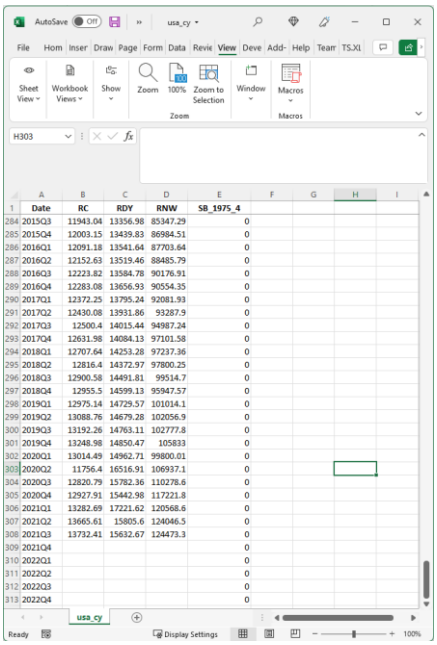
1) Drag/drop a file in the main EViews window (this is the easiest method).
2) Click **File → Open → Foreign Data as Workfile**.
3) Copy/Paste as New Workfile in the EViews desktop.

To illustrate, we will now create a workfile from an Excel file containing time-series data on USA household consumption expenditure. The file is called ***usa_cy.xlsx***, and contains the following 4 (quarterly) series for the period 1947:1 2021:3:

1. Real consumption expenditure (**rc**)
2. Real disposable income (**rdy**)
3. Real net worth (**rnw**)
4. Structural break dummy for the period 1947:1 – 1975:4 (sb_1975_4). The idea is to capture a permanent change in household consumption expenditure around 1975:4

The data in Excel is shown in the following screen image:



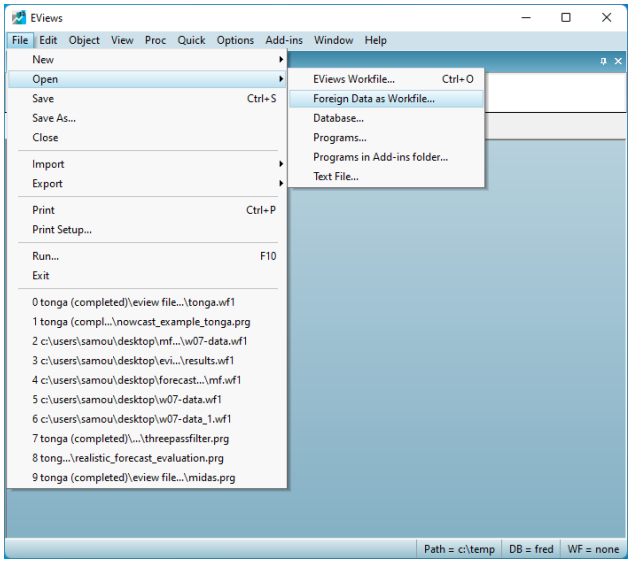


[3] It is strongly recommended that you start a new session of EViews before importing a foreign dataset. This is to avoid accidentally overriding existing data.

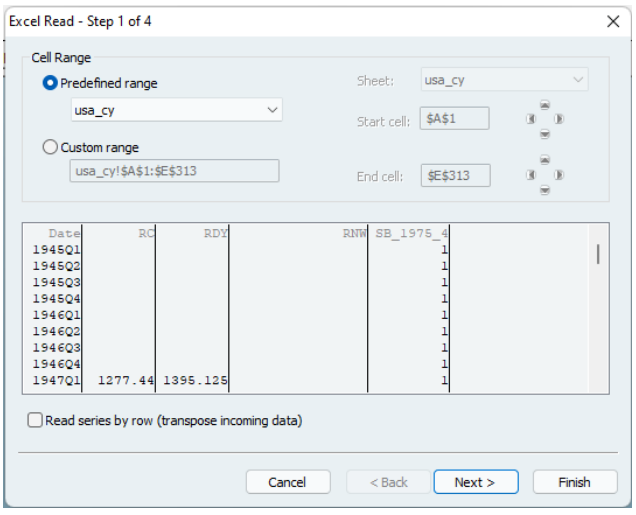
Notice that each of these variables is stored in a single column of the Excel worksheet, while the first column shows dates. The four variables define a time-series matrix with a specific start and end date.

For this dataset, time starts in 1945, quarter 1 (1945:1), and ends in 2022, quarter 4 (2022:4). The quarterly data for real consumption and real disposable income starts in 1947:1 and ends in 2021:3, while that for real net worth commences in 1952:1 and ends in 2021:3. Data on these variables outside of these date ranges are regarded as missing variables (or "NA" values in EViews language). The variable designed to capture a possible structural change in consumption behavior from 1975:4 onwards, **sb_1975_4**, starts in 1945:1 and ends in 2022:4. This variable does not have any missing values.

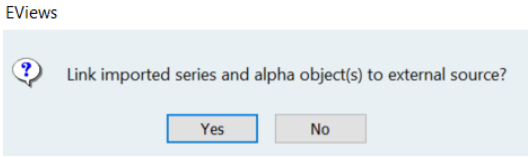
You can import the data in this spreadsheet into EViews several ways. The first is to click **File → Open → Foreign Data as Workfile**, namely:



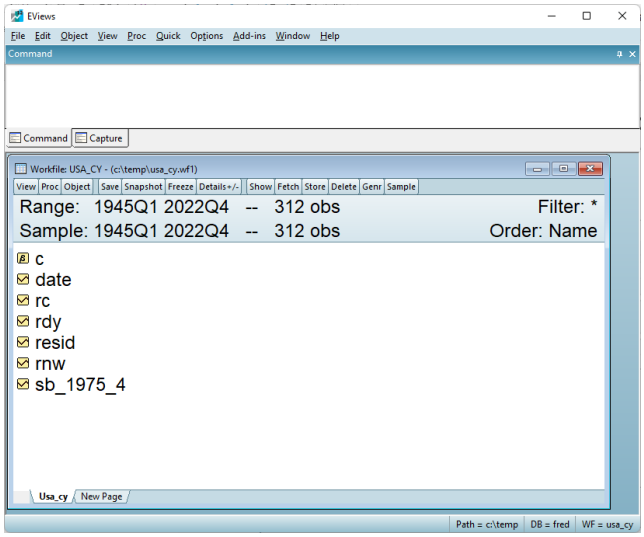
And then select the directory where the Excel sheet **usa_cy.xlsx** is saved and open the file. The following Excel read dialog box appears, with pre-populated entries assigned by EViews:



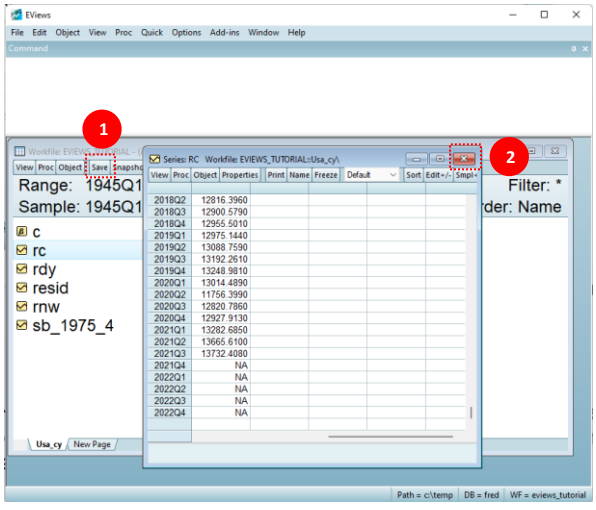
Clicking the "Finish" button instructs EViews to import the data from Excel.[4] A pop up window appears asking you whether "link imported series and alpha object to external source?". Click on "No"[5].



The result is a workfile called "usa_cy", with a single pagefile also called "usa_cy". The frequency of the pagefile is quarterly, and its range is 1945:1-2022:4.



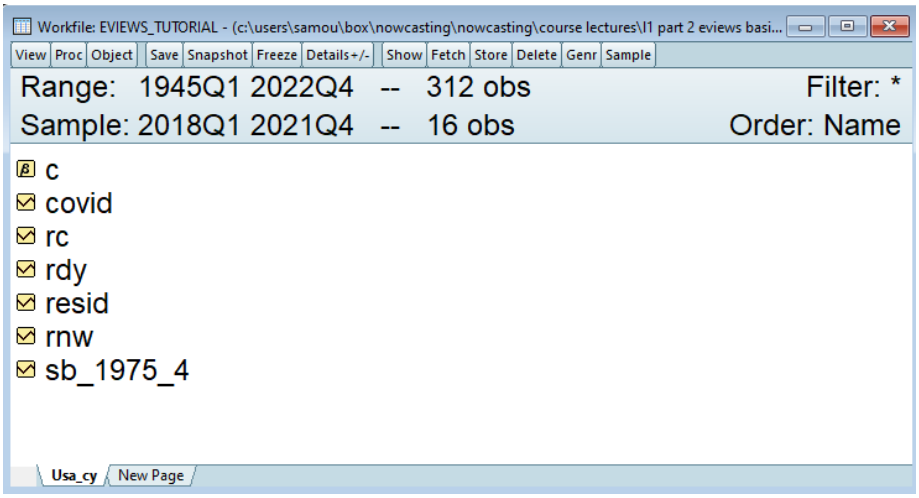
Double-clicking on the "**rc"** series object provides a spreadsheet view of the data attached to the series object, namely:



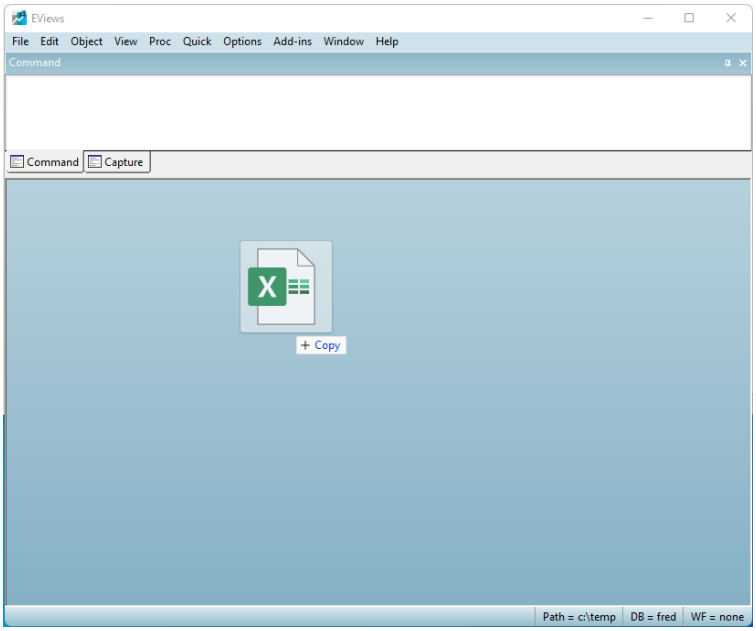
[4] The "Next>" button brings up further dialog boxes that allow you to change the default behavior of the import routine.

[5] If click "Yes", series imported to EViews will update automatically when updates occurred in the foreign sources. If click "No", series will not be "copied" and will not synchronize updates in foreign sources.

Lastly, click on the "Save" button of the workfile view to save your work (1). You can close the workfile view by clicking on the upper, red X button in the workfile view (2).

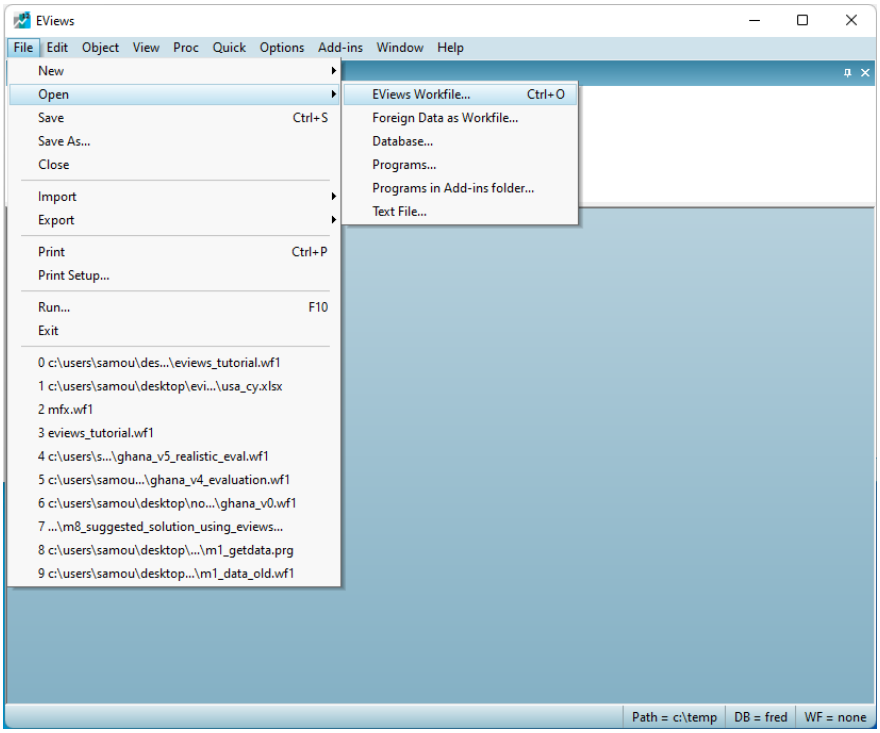


Another way to achieve the same result is to drag and drop the Excel sheet **usa_cy.xlsx** into the command output area of EViews. This will cause EViews to open the Excel file and bring up EView's Excel read dialog box as before.

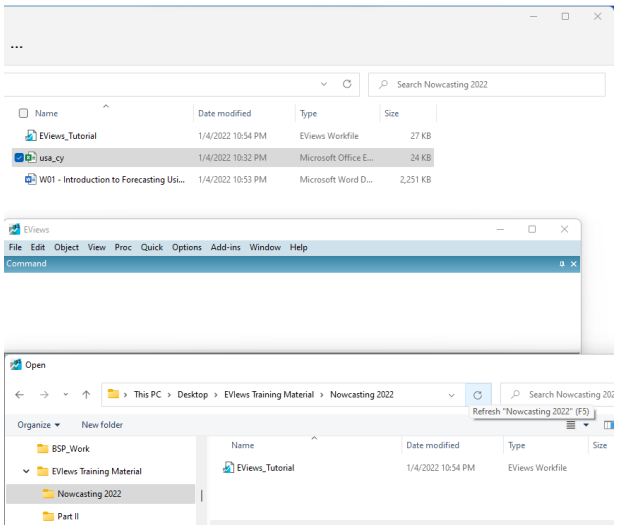


6. OPENING AN EXISTING WORKFILE

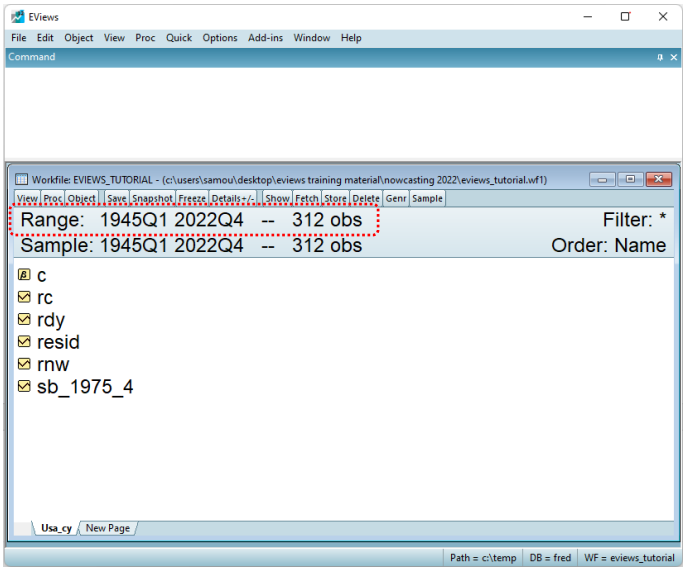
Use the **File** menu (or CTRL+O) to open an existing workfile, specifically "File → Open → EViews Workfile (CTRL+O)". Select the EViews workfile to be opened, namely **EViews_Tutorial.WF1**, using the resulting file dialog box:



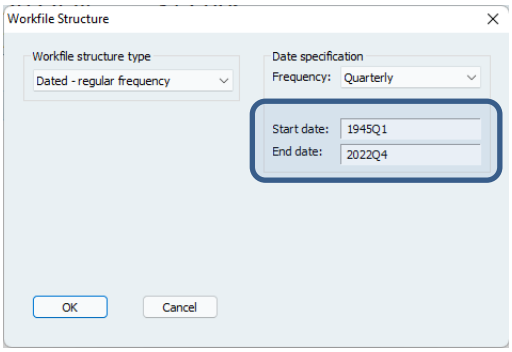
The resulting file dialog box is:



Opening the EViews file reveals the objects saved in the workfile and enables you to start analyzing your data, which will typically be stored in separate series objects. We will start working with this file from here.



Lastly, there may be occasions where you need to change the workfile range after you have created the workfile. You can do this by double-clicking on the existing range indicator (e.g., "Range: 1945Q1 2022Q4") and changing the start date and/or end date (highlighted) in the resulting dialog box, namely:



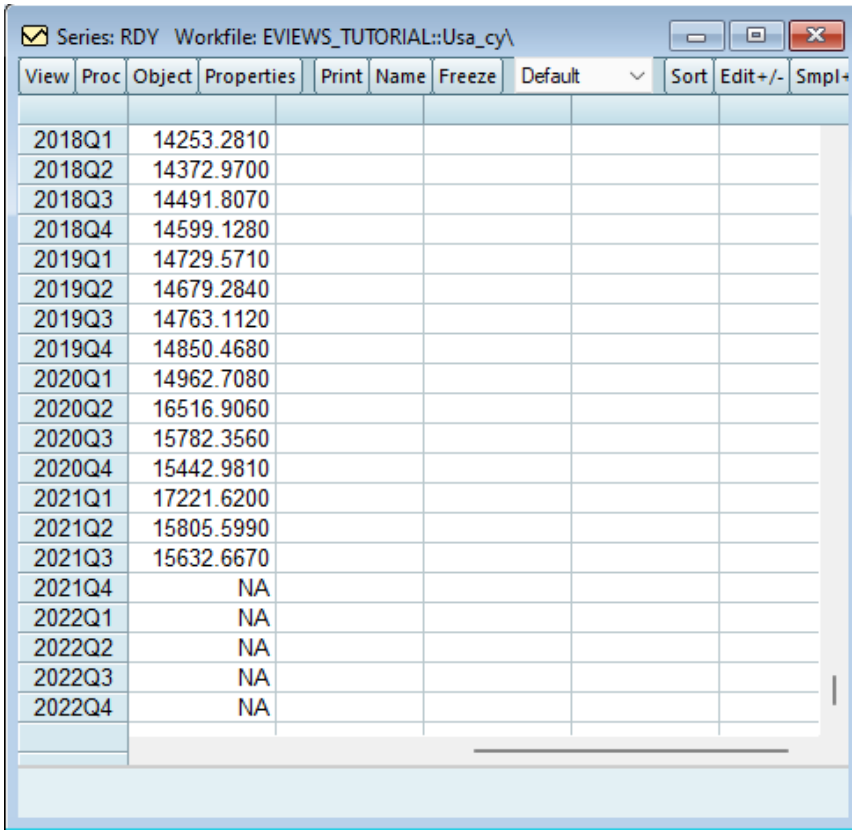
The existing objects in the workfile will remain the same. Only their data capacity will change to reflect the new range.

7. WORKING WITH SERIES

There are many ways to inspect the data stored in a series object.

1) Mouse

As shown previously, you can double-click on the series object, e.g."**rdy**". The following screen will appear after you scroll to the end of the series:



2018Q1	14253.2810
2018Q2	14372.9700
2018Q3	14491.8070
2018Q4	14599.1280
2019Q1	14729.5710
2019Q2	14679.2840
2019Q3	14763.1120
2019Q4	14850.4680
2020Q1	14962.7080
2020Q2	16516.9060
2020Q3	15782.3560
2020Q4	15442.9810
2021Q1	17221.6200
2021Q2	15805.5990
2021Q3	15632.6670
2021Q4	NA
2022Q1	NA
2022Q2	NA
2022Q3	NA
2022Q4	NA

Click on the upper, red X button () to close this view.

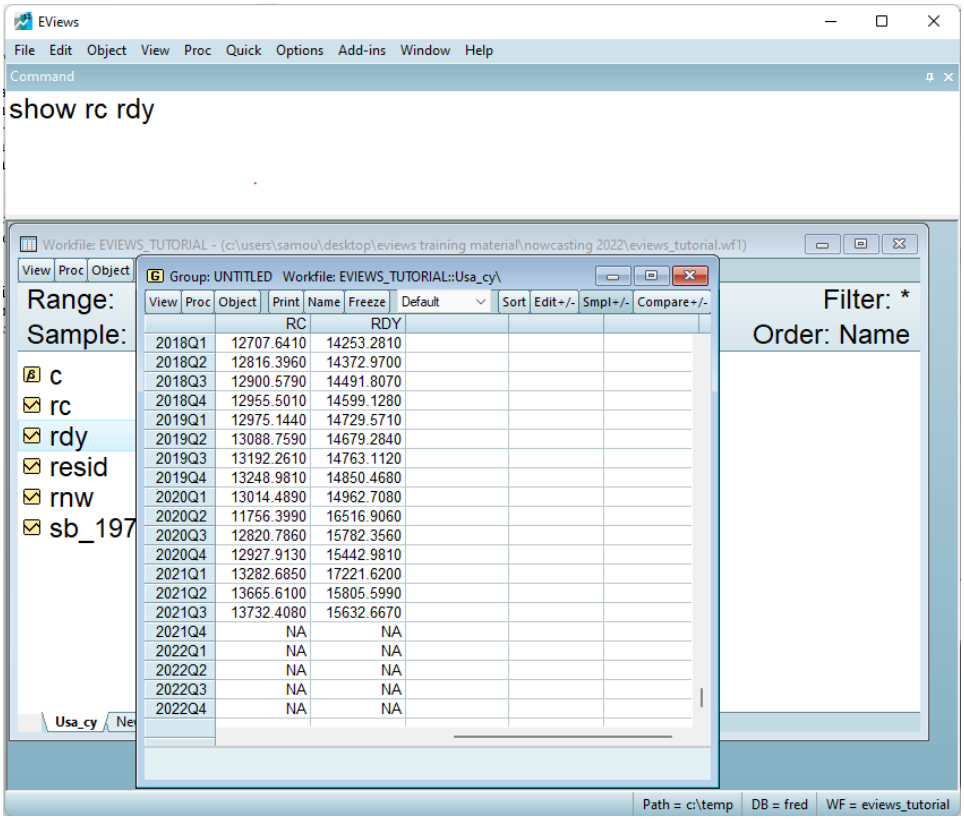
2) Command Line:

The same outcome can be achieved by typing "**show rdy**" in the command line.

After you open the series window, you can easily change the data stored in the series. But before you can do so, you need to click on the **Edit+/-** button, which acts as an on/off switch for the EViews data editor. Experiment by turning on editing and make changes to the existing data values. The changes you make are saved once you close the series viewer/spreadsheet or click on the **Edit+/-** button again to turn off the editor.

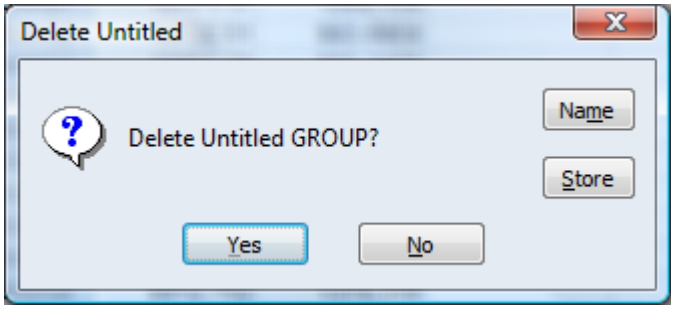
To work on a collection of series, select the series you are interested in working with using the "**Ctrl Key + Mouse Click"** combination, then right-click and **Open → As Group**

Alternatively use **Object → New Object → Group** and then click **OK**. Then type in the name of the series you want to study as a group. Lastly, from the command line, you can also type, for example, **show rc rdy.** This produces an untitled group that can be saved in the workfile:



You will be prompted to create a permanent group containing "**rc**" and "**rdy**" when you close the "**UNTITLED**" group view for the first time (or close the workfile itself). Saving the group view will use *pointers* to the series objects rather than making copies of the series themselves. Consequently, any changes you make to the series will be reflected in the group view the next time it is opened.

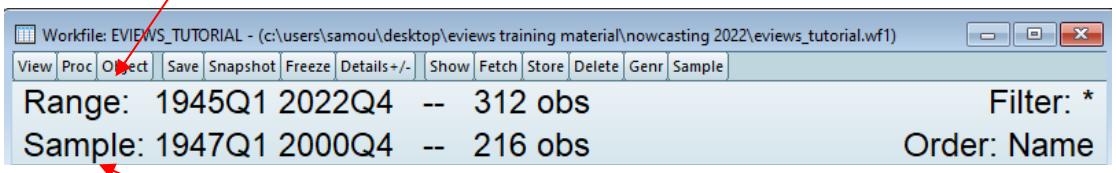
This is the dialog box that allows you to name the untitled group and save it permanently in the work file. Alternatively, you can discard the group by clicking on "Yes" which is the default setting (careful).



8. EVIEWS SAMPLES

Samples in EViews are sets (usually subsets) of observations in a workfile. They control the range over which statistical operations (e.g., regression) are performed. Samples may be specified over ranges of observations and "if" conditions.

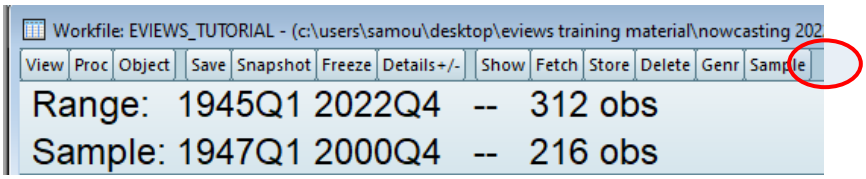
The workfile range is 1945Q1 to 2022Q4, a total of 312 observations.



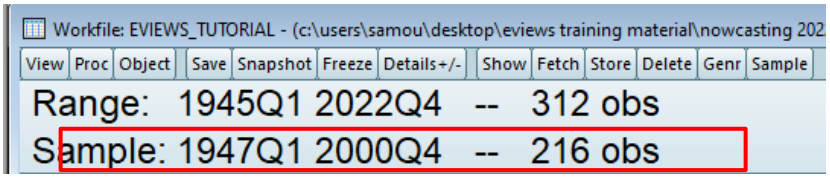
The active sample is 1947Q1 to 2000Q4, a subset of the workfile range, or 216 observations.

There are several ways to change/set the workfile sample in EViews:

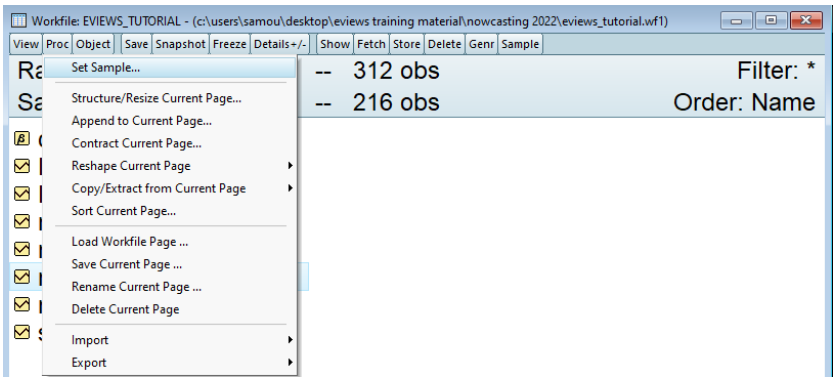
1) Click the **Sample** button in the workfile toolbar.



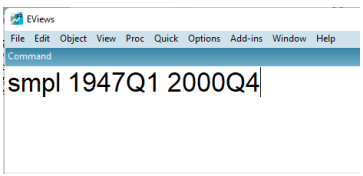
2) Double click on the **Sample** display in the workfile window.



3) Select **Proc/Set Sample** from workfile menu.



4) Type **"smpl"** in the command window.



Note that sample specifications are always entered in pairs, e.g.,1947Q1 2000Q4.

EViews has special keywords that make it easier/quicker to change the sample to useful subsets of the workfile range:

Function	Command	Description
@all	smpl @all	Refers to the workfile range (includes all observations)
@first @last	smpl @first @last	Also refers to the workfile range (all obs. from first to last**)**
@first @first	smpl @first @first	Includes the first observation only
@last @last	smpl @last @last	Includes the last observation only

Sample ranges can be defined by using mathematical expressions. This is normally used to define a fixed-width window of observations.

Function	Command	Description
@first+12 @last	smpl @first+12 @last	Excludes the first 12 observations
@first @first+11	smpl @first @first+11	Captures only the first 12 observations
@first @last-15	Smpl @first @last-15	Excludes the last 15 observations
@last-14 @last	smpl @last-14 @last	Captures only the last 15 observations
1960m3 1960m3+11	smpl 1960m3 1960m3+11	Defines a sample with 12 observations starting in *March 1960 – Feb 1961*

Section 2: Creating New Series from Existing Series

EViews contains a powerful equation parser that allows you create new series from existing series in the active page you are working with.

1. USING THE COMMAND LINE TO INITIATE CALCULATION

You can use the "**genr**" (or equivalently the "**series**") command to create new series from existing ones. For example, to take the logs of **rc** and **rdy** in **usa_cy**, you can issue the following command:

genr l_rc = log(rc)
genr l_rdy = log(rdy),

And to calculate percentage change, use :

genr p_rc = (((rc-rc(-1))/rc(-1)))*100

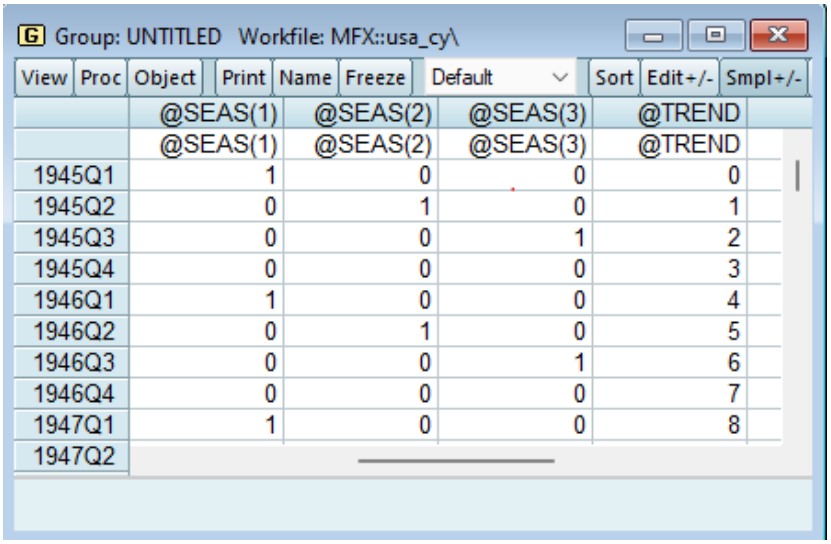
EViews understands the standard algebraic operators: + - * / ^, and many more (see next image). EViews ensures that the observations of the series referenced in the **genr** statement are aligned correctly by date before performing any operations. Note that all the series in a **genr** statement must have the same frequency.

EViews Operators: Genr/Series Statement

Expression	Operator	Description
+	add	x+y adds the contents of X and Y.
-	subtract	x-y subtracts the contents of Y from X.
*	multiply	x*y multiplies the contents of X by Y.
/	divide	x/y divides the contents of X by Y.
^	raise to the power	x^y raises X to the power of Y.
>	greater than	x>y takes the value 1 if X exceeds Y, and 0 otherwise.
<	less than	x<y takes the value 1 if Y exceeds X, and 0 otherwise.
=	equal to	x=y takes the value 1 if X and Y are equal, and 0 otherwise.
<>	not equal to	x<>y takes the value 1 if X and Y are not equal, and 0 if they are equal.
<=	less than or equal to	x<=y takes the value 1 if X does not exceed Y, and 0 otherwise.
>=	greater than or equal to	x>=y takes the value 1 if Y does not exceed X, and 0 otherwise.
and	logical and	x and y takes the value 1 if both X and Y are nonzero, and 0 otherwise.
or	logical or	x or y takes the value 1 if either X or Y is nonzero, and 0 otherwise.

EViews also has many built-in functions that allow you to perform data manipulation/create series. EViews functions are typically denoted by the symbol "**@**".

For example, **@seas(n),** n = 1, 2, 3 …**,** creates a seasonal dummy that is 1 for the Nth period (e.g., quarter) of the year, and 0 elsewhere. The function **@trend** creates a time trend variable that starts with 0 for the first period of the workfile range.



	@SEAS(1)	@SEAS(2)	@SEAS(3)	@TREND
	@SEAS(1)	@SEAS(2)	@SEAS(3)	@TREND
1945Q1	1	0	0	0
1945Q2	0	1	0	1
1945Q3	0	0	1	2
1945Q4	0	0	0	3
1946Q1	1	0	0	4
1946Q2	0	1	0	5
1946Q3	0	0	1	6
1946Q4	0	0	0	7
1947Q1	1	0	0	8
1947Q2				

You can also calculate cumulative statistics using functions like "@cumsum" or moving average statistics with "@movsum". For each function, you have the option to control the sample (i.e., @sum(s1, "1 3") yields the sum of s1 for only the first three elements). You may also generate expression for series within the group using @ functions like "@rsum(group_name)".

2. LEADS AND LAGS IN EVIEWS

It is easy to work with lags/leads and other time series functions in EViews. You do not need to generate series of lags/leads in many places in EViews; simply write the command for them when needed (i.e., when estimating a regression).

Common Commands/Functions

Function	Description
rc(-4)	Denotes the 4th lag of the **rc** series
rc(2)	Denotes the 2nd lead of the **rc** series
rc(-1 to -4)	Specifies all **rc** lags from 1 to 4
rc(to -5) OR rc(0 to -5)	Specifies all **rc** lags from 0 to -5
series con = @lag(rc,3)	Generates series **con**, as the 3rd lag of **rc**
series y =@lag((rc-rdy)/rc,4)	Generates series **y**, as the 4th lag of the transformation **(rc-rdy)/rc**

EViews has several build-in functions for working with **differenced data** (levels and logs).

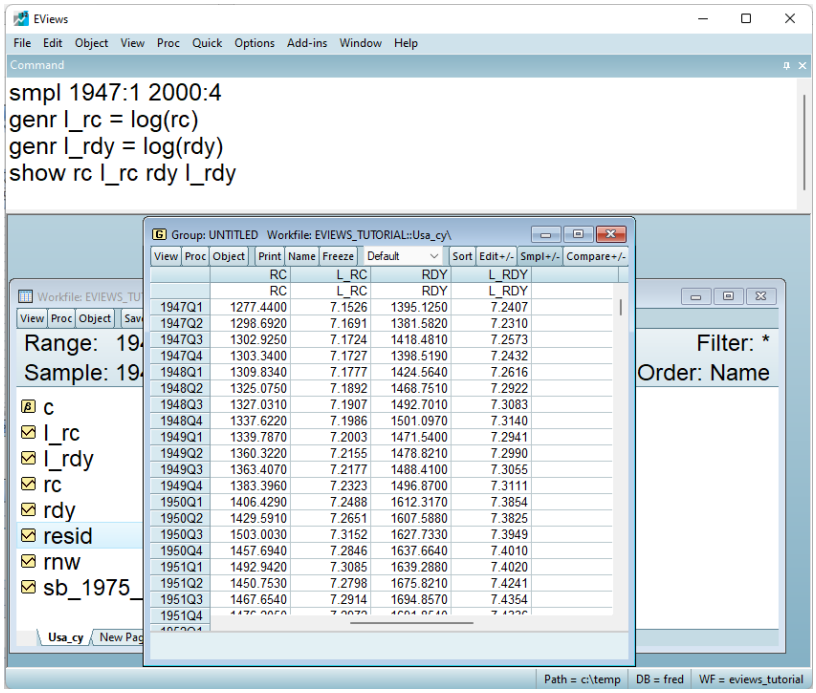
Function	Description
d(rc) = rc - rc(-1)	Takes the first difference of the **rc** series
dlog(rc) = log(rc) – log(rc(-1))	Takes the first difference of **log(rc)** series

It also has functions to calculate **percentage changes**:

Function	Description
@pc(rc)	calculates the one-period percent change in **rc** (in percent)
@pch(rc)	calculates the one-period percent change in **rc** (in decimal)
@pca(rc)	calculates the one-period ***annualized*** percent change in **rc** (in percent) $(1+@pch(x))^n-1$ (where n=4 for quarterly data, n=12 for monthly, ect)
@pcha(gdp)	calculates the one-period ***annualized*** percent change in GDP (in decimal)
@pcy(gdp)	calculates the one-year percent change (in percent)
@pchy(gdp)	calculates the one-year percent change (in decimal)

Please consult the EViews manual for complete details of the standard algebraic operators and the "@" functions.

<u>It is very important to note</u> that the **genr** command operates only the data values in the active sample of the workfile, and that the active sample could be smaller than the actual range of the workfile. A common error made when using EViews is to use the incorrect sample when issuing **genr** commands.



For example, preceding the "**genr**" command with the command

smpl 1947:1 2000:4

forces the "**genr**" command (and all subsequent commands) to operate over 1947:1 2000:4 (i.e., a subset of the entire sample).

You can restore the default behavior by issuing

smpl @all

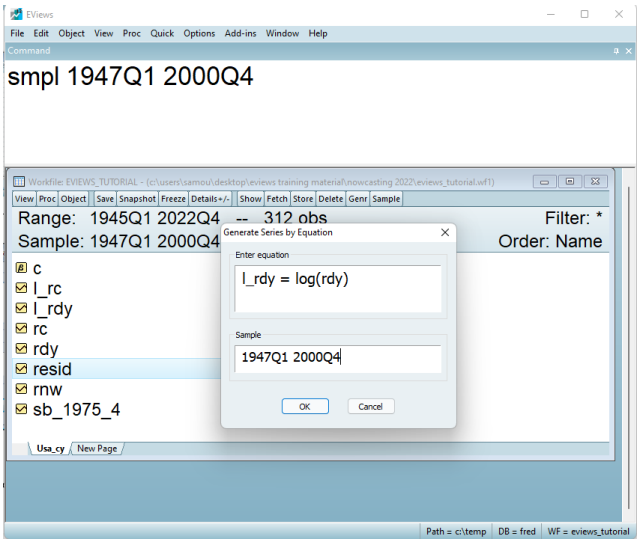
after which the **genr** command will operate over the range of the workfile (i.e., 1945:1 2022:4).

A 0/1 dummy variable can be easily created using genr and smpl statements. For example, suppose we wish to create the structural break dummy sb_1975_4, which is equal to 1 for the period 1945:1 to 1975:4, and 0 otherwise. The following commands will produce this variable:

```
smpl @all
genr sb_1975_4 = 0
smpl @first 1975:4
genr sb_1975_4 = 1
```

3. GENERATING SERIES USING THE MOUSE

Click on the "**genr**" button on the workfile menu and then enter the equation, here l_rdy = log(rdy), together with the desired sample in the sample dialog box immediately below the "Enter equation" field:

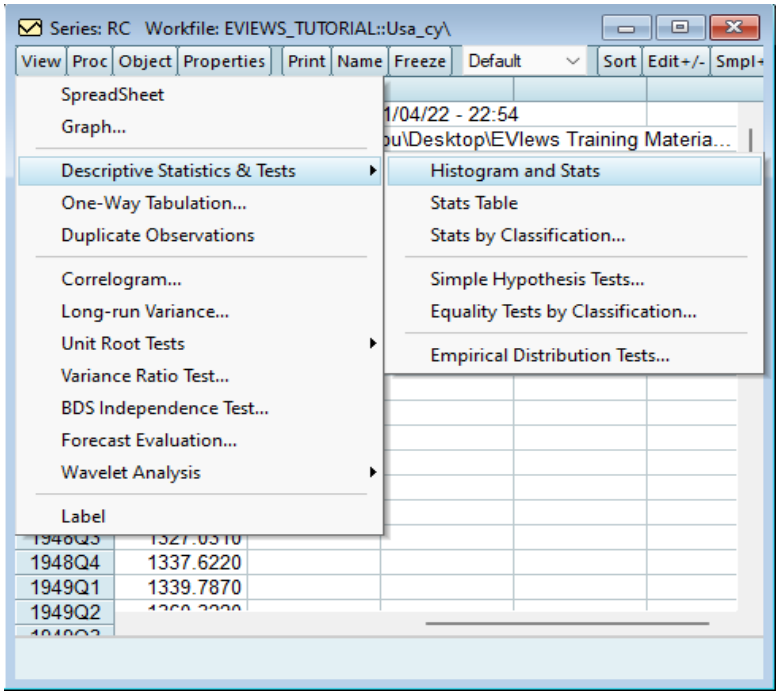


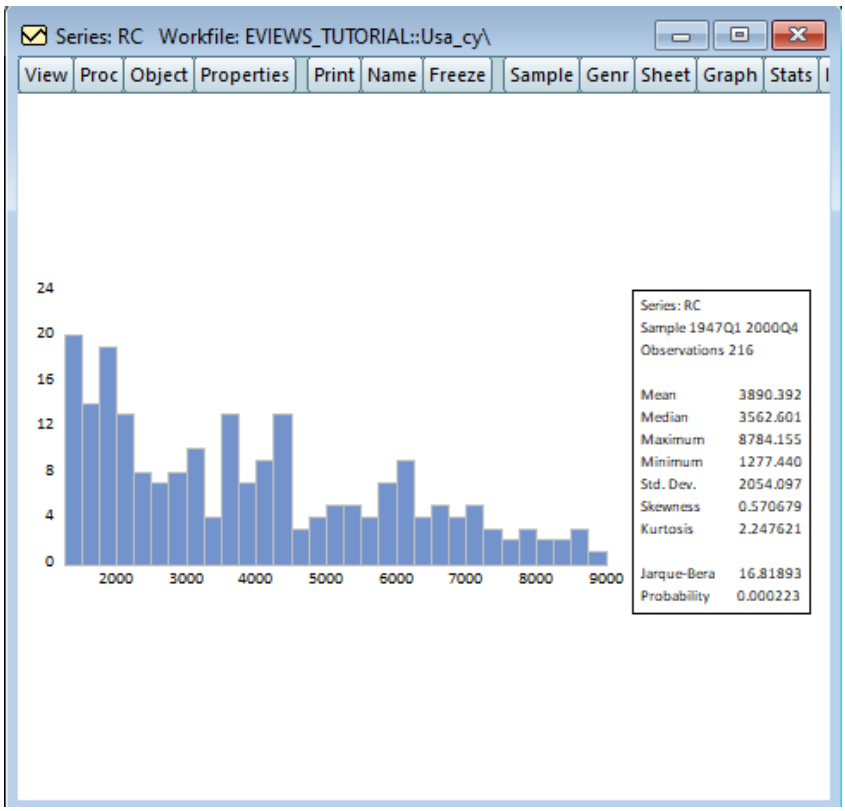
Section 3: Descriptive Statistics

Select the series called "**rc"** then open it in separate window to examine the data.

1. HISTOGRAM AND STATS

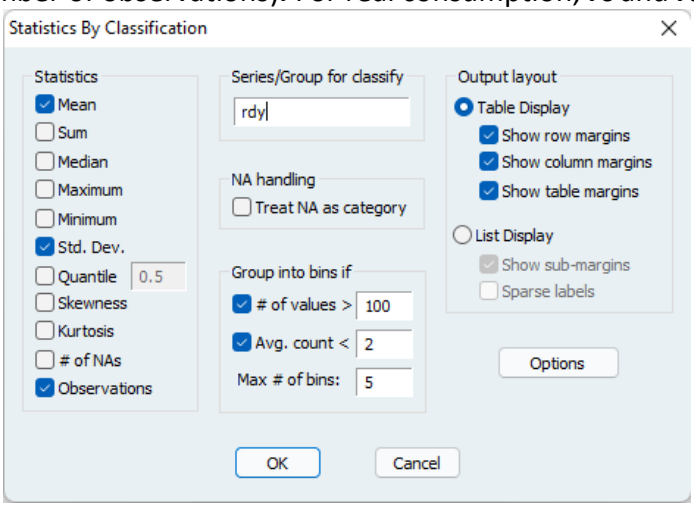
View → Descriptive Statistics and Tests → Histogram and Stats



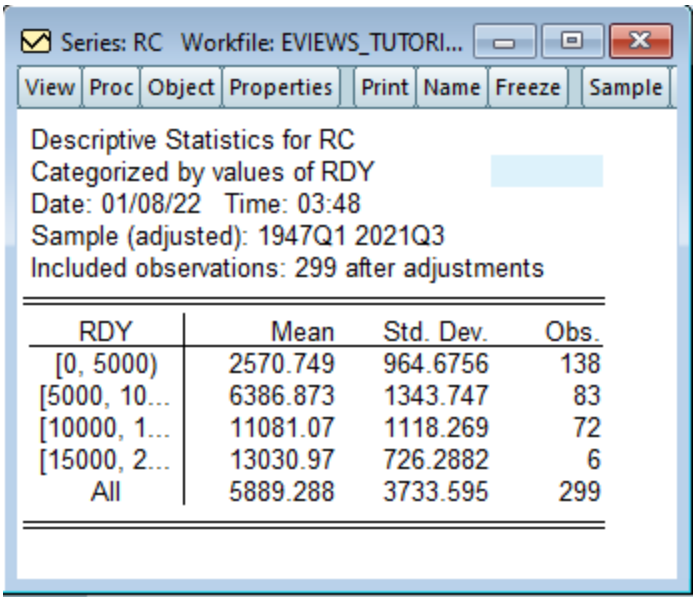


2. STATS BY CLASSIFICATION

View → Descriptive Stats → Stats by Classification, then select the variables to be used for classification along with the descriptive statistics you need (for example, mean, standard deviation and the number of observations)**.** For real consumption, **rc** and **rdy**, say:



yields the following output:



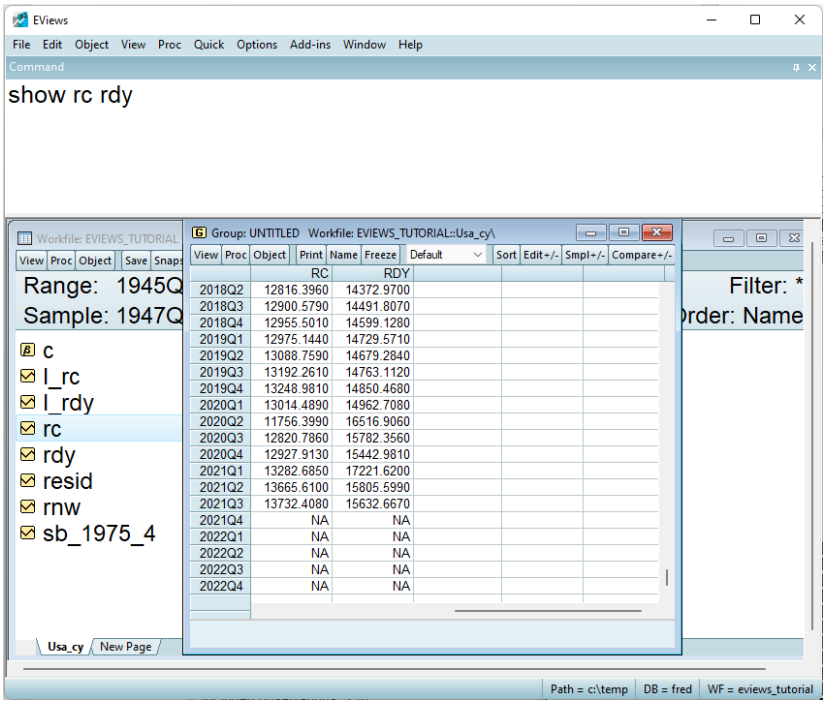
RDY	Mean	Std. Dev.	Obs.
[0, 5000)	2570.749	964.6756	138
[5000, 10...	6386.873	1343.747	83
[10000, 1...	11081.07	1118.269	72
[15000, 2...	13030.97	726.2882	6
All	5889.288	3733.595	299

Implying, for example, that the average of real consumption, rc, is 2570.749 for rdy < 5000 based on 138 valid observations. The unconditional mean of rc is 5889.288 for 299 observations.

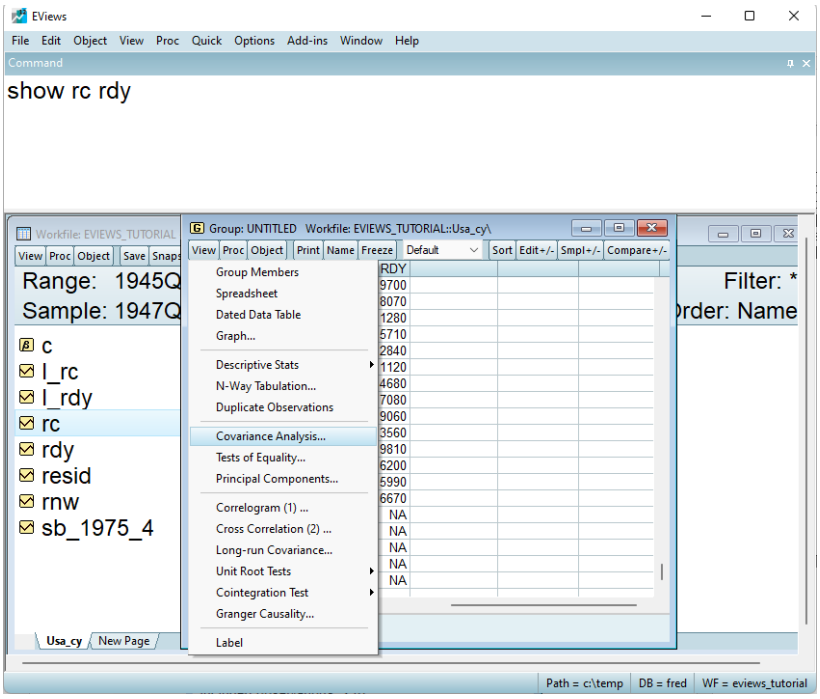
3. COVARIANCE ANALYSIS

Having selected **rc** and **rdy** for joint analysis, multivariate summary statistics can be easily produced using **View → Covariance Analysis.** The first step is to use the **show** command to create a temporary group by selecting **rc** and **rdy** together, viz:

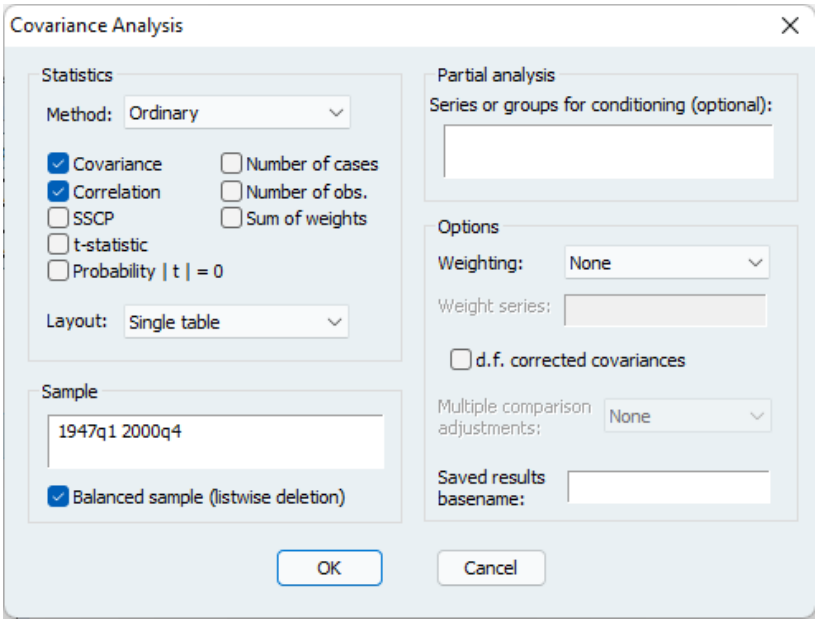
show rc rdy



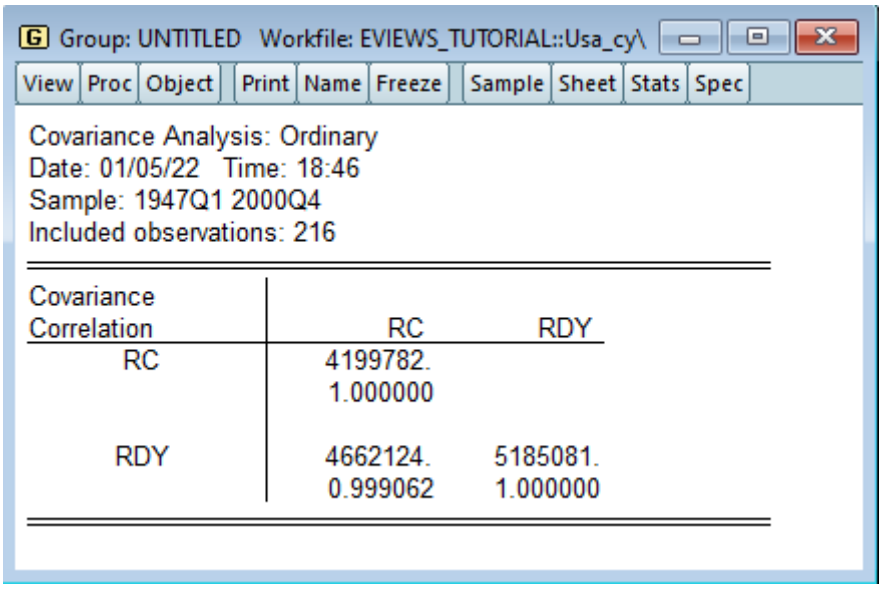
Then click on **View → Covariance Analysis**



to obtain the following dialog box that allows you to select various multivariate statistics and control the way the descriptive statistics are calculated and presented:



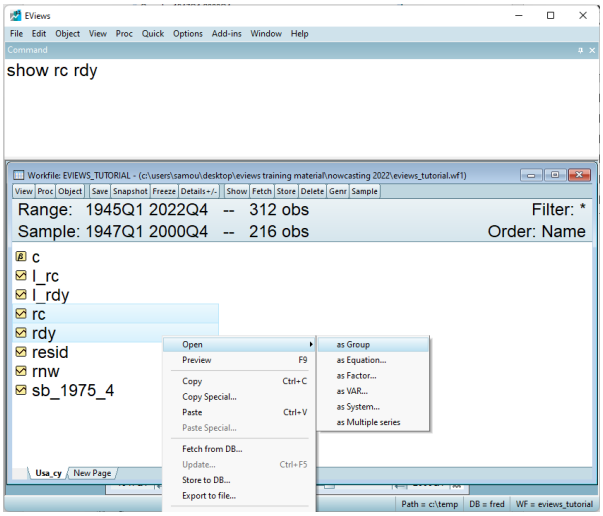
For the options selected above, pressing OK yields the following results:

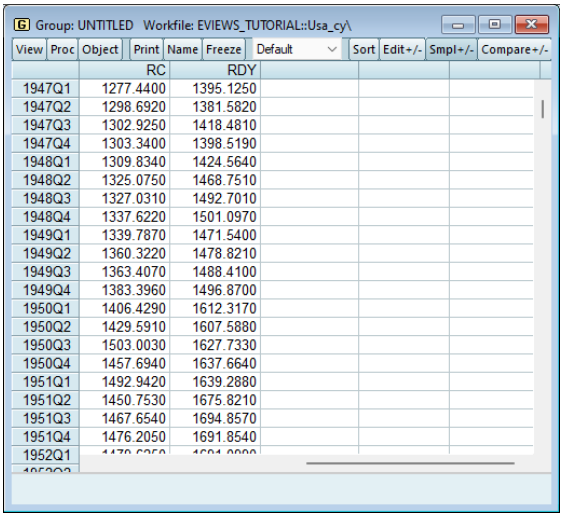


They indicate a very high positive correlation of 0.99 between **rc** and **rdy**, as to be expected.

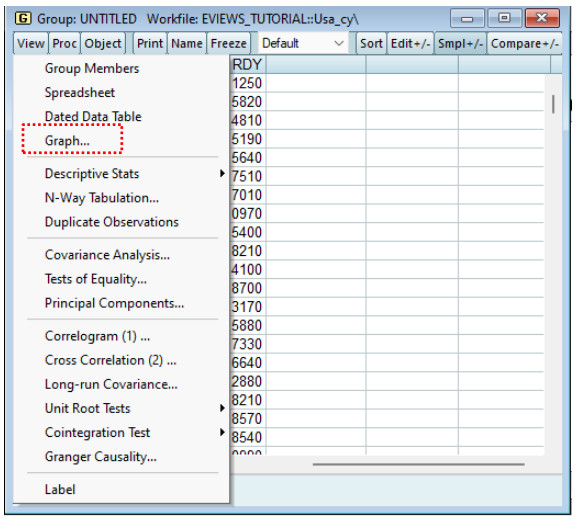
Section 4: Graphical representations

Using the mouse and holding the CTRL key down, select **rc** and **rdy**, right click, and then open them as a group.

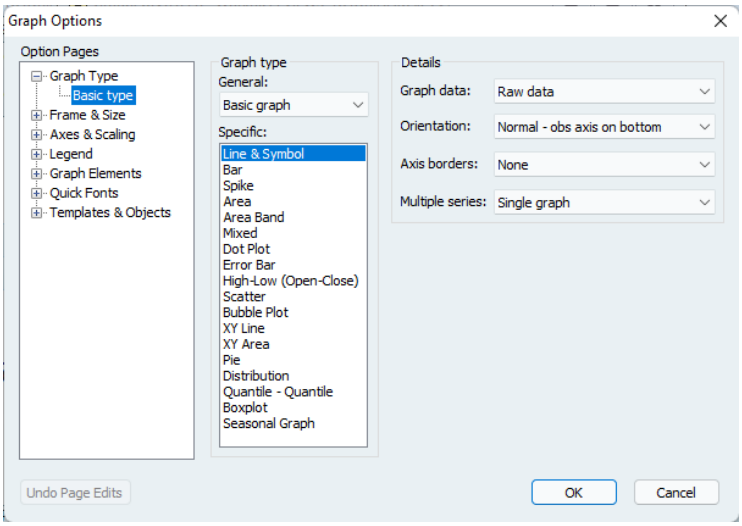




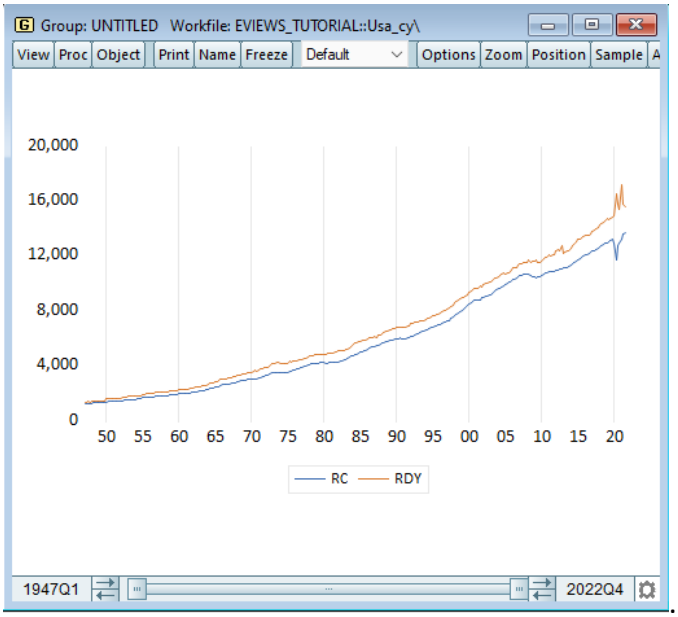
Then View→Graph



1. SINGLE GRAPH



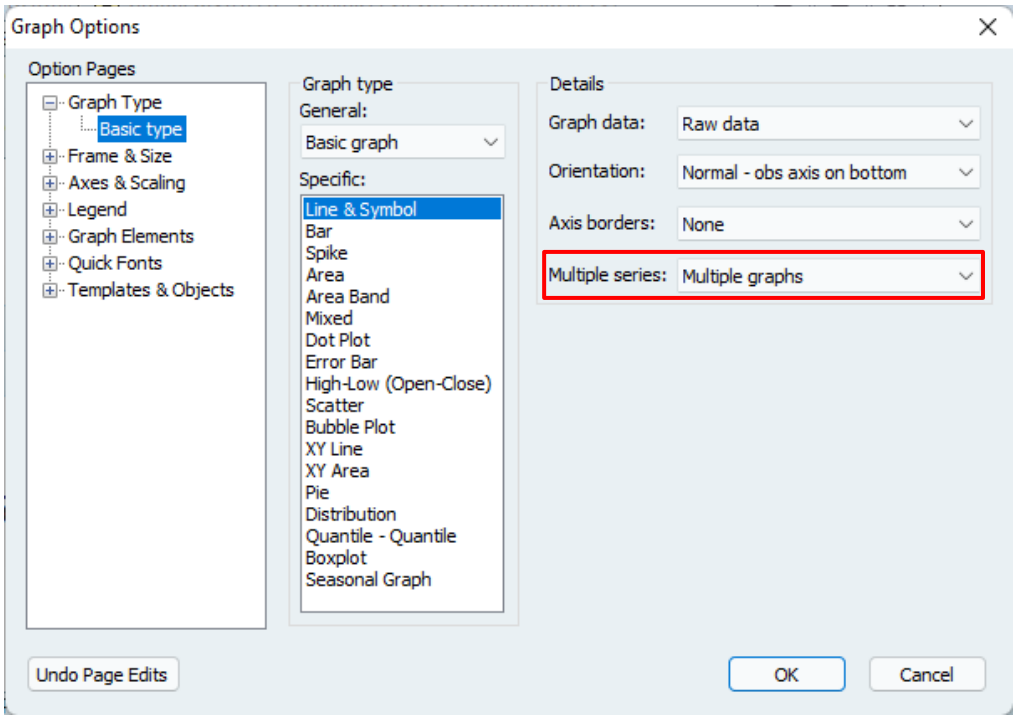
Make sure the settings correspond to those shown above and click "**OK**" to view the two series on a single (line) graph



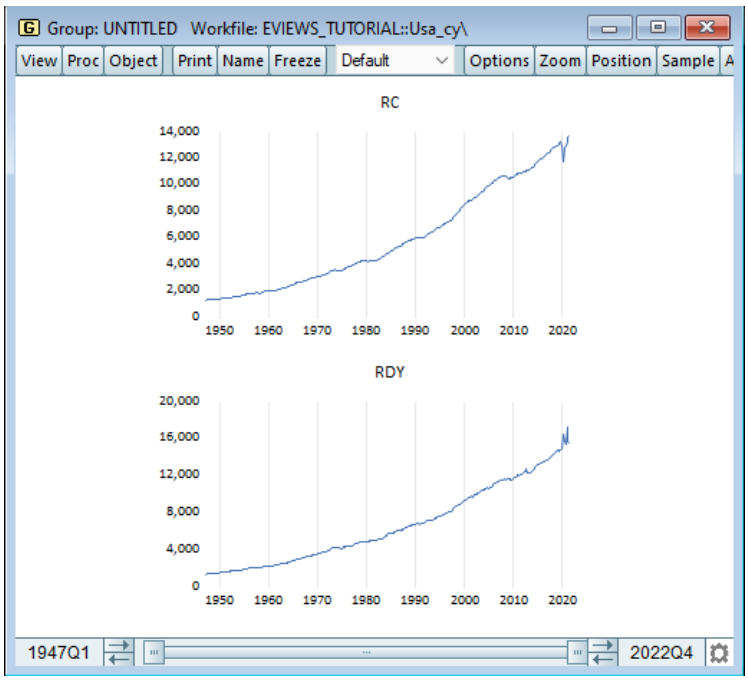
Note the availability of the sample slide bar at the bottom of the graph, which allows you to focus on a particular sample of the data. You can adjust the sample by clicking on the left and right arrows of the sample slide bar.

2. MULTIPLE GRAPHS

You can optionally elect to graph each series in a separate graph. To do so, click on the "Multiple Series:" drop down box and select "Multiple graphs"

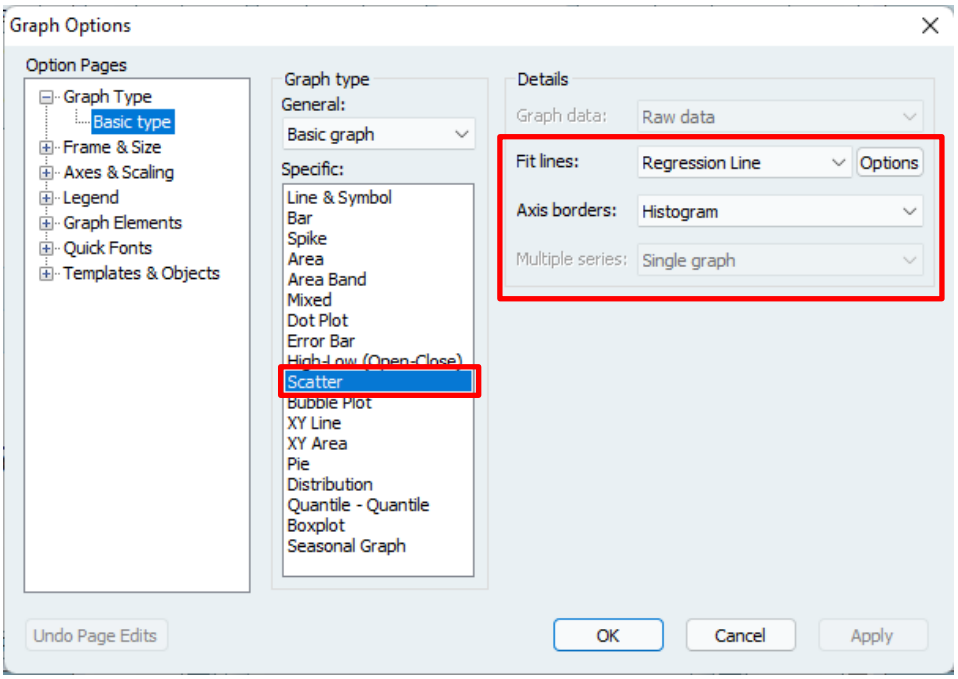


The resulting output is given below:

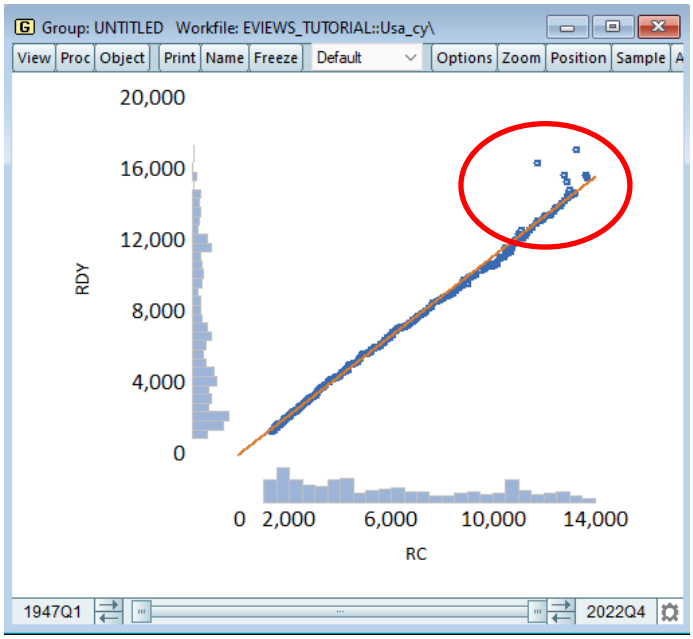


3. OTHER GRAPH TYPES

EViews supports several graph types



including scatter plots:



Notice the almost linear relationship between consumption and real disposable income. The observations that are not near the linear trend line all occurred during the COVID pandemic period (2020-2021).[6] They are all within the red circle in the XY plot.

Section 5: Estimating a Regression Line Using Ordinary Least Squares (OLS)

We now turn to estimating a basic regression using EViews. To illustrate, we shall estimate the following, conventional, consumption function for the USA:

log(rc) c log(rdy) log(rnw) log(rc(-1)) sb_1975_4 covid

where covid is a 0/1 dummy variable to allow for the severe effects of COVID-19 on US household consumption behavior. The covid dummy variable has a value of 1 for the 2020Q1-2021Q4 period, and 0 otherwise. It was created using the following command sequence:

```
smpl @all
series covid = 0
smpl 2020Q1 2021Q4
series covid = 1
```

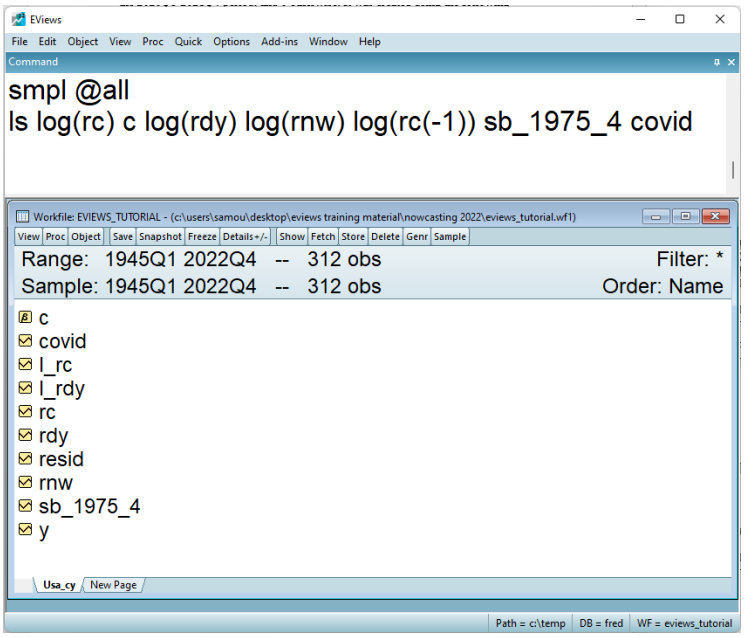
Notice that the regression includes a lagged dependent variable, as required by the permanent income hypothesis and standard error-correction models of consumption.

Single equation regression estimation in EViews is performed using the ***Equation Object.*** There are several ways to create a simple OLS Equation Object:

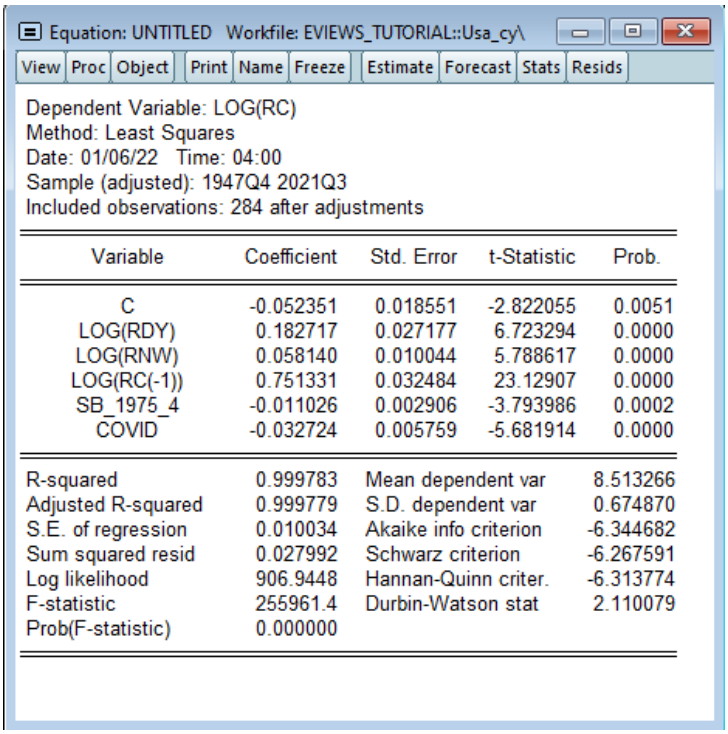
[6] A detailed tutorial on basic EViews graphs can be downloaded from Graphs (eviews.com)

1. USING THE COMMAND LINE TO CREATE EQUATION OBJECT

Use the "**ls**" command to estimate a regression, viz:



The regression results, using ordinary least squares, are



Equation: UNTITLED Workfile: EVIEWS_TUTORIAL::Usa_cy\

View | Proc | Object | Print | Name | Freeze | Estimate | Forecast | Stats | Resids

Dependent Variable: LOG(RC)
Method: Least Squares
Date: 01/06/22 Time: 04:00
Sample (adjusted): 1947Q4 2021Q3
Included observations: 284 after adjustments

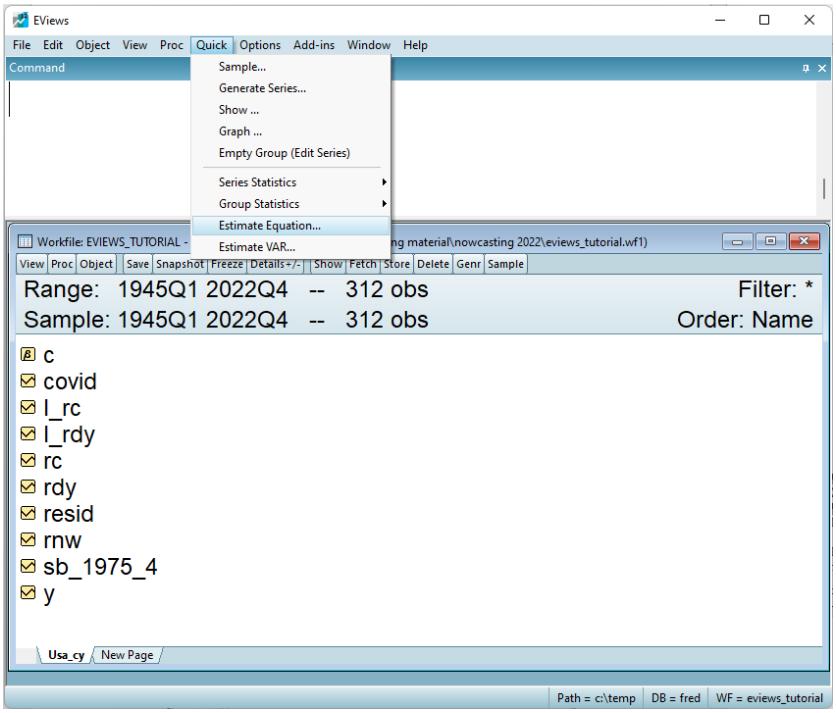
Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	-0.052351	0.018551	-2.822055	0.0051
LOG(RDY)	0.182717	0.027177	6.723294	0.0000
LOG(RNW)	0.058140	0.010044	5.788617	0.0000
LOG(RC(-1))	0.751331	0.032484	23.12907	0.0000
SB_1975_4	-0.011026	0.002906	-3.793986	0.0002
COVID	-0.032724	0.005759	-5.681914	0.0000

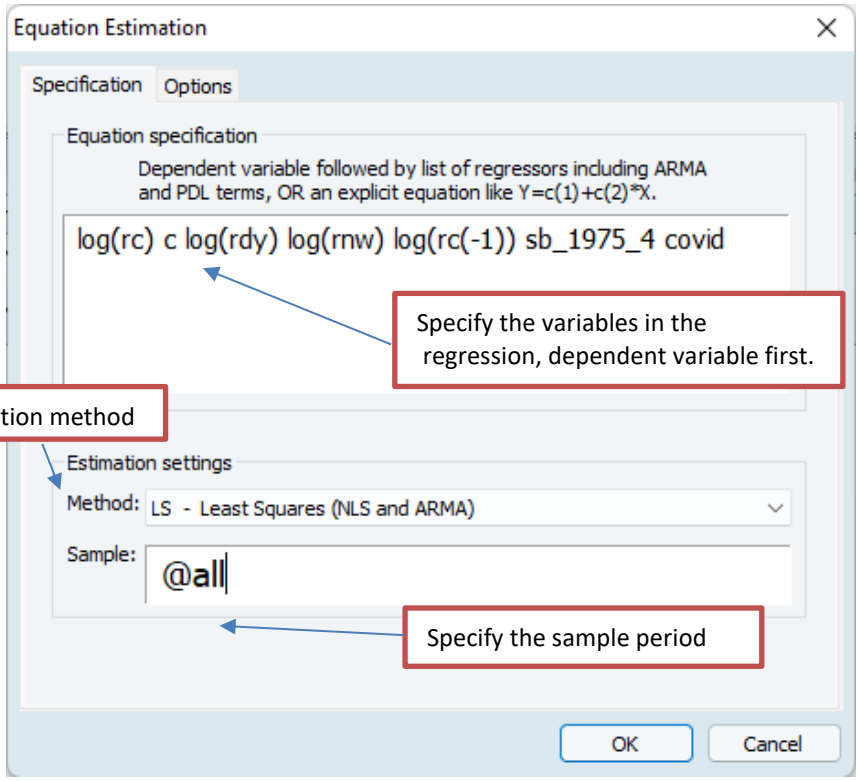
R-squared	0.999783	Mean dependent var	8.513266
Adjusted R-squared	0.999779	S.D. dependent var	0.674870
S.E. of regression	0.010034	Akaike info criterion	-6.344682
Sum squared resid	0.027992	Schwarz criterion	-6.267591
Log likelihood	906.9448	Hannan-Quinn criter.	-6.313774
F-statistic	255961.4	Durbin-Watson stat	2.110079
Prob(F-statistic)	0.000000		

Note that you can use data expressions (e.g**., log(rc)**) directly in the equation box to estimate a regression without having to first create these series. These can be specified directly in the equation box.

2. USE "QUICK" TO CREATE EQUATION OBJECT

Another way to estimate a regression is to use **Quick → Estimate Equation**

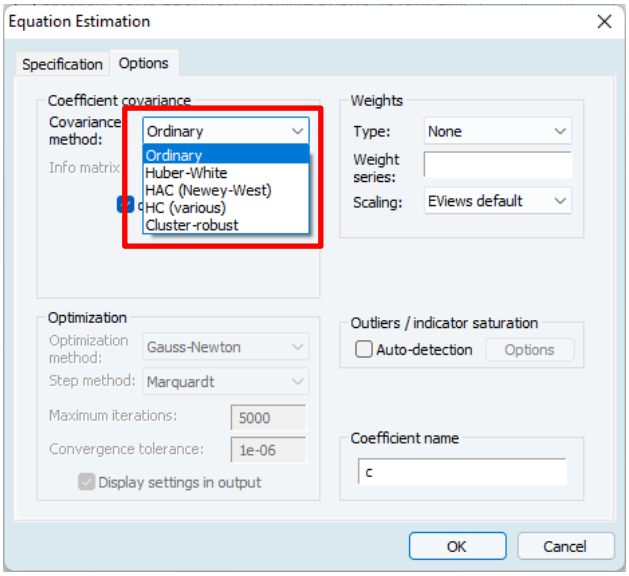




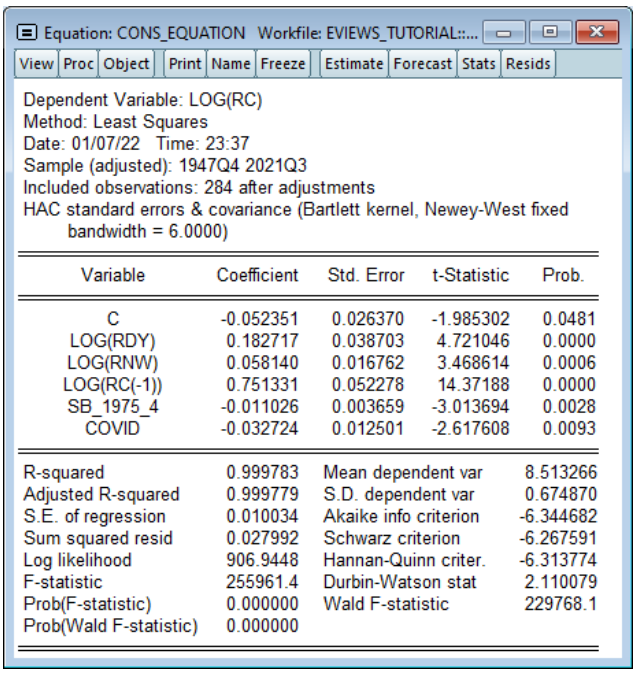
Then enter the variables in the regression in the appropriate dialog boxes, as shown above.

You can also control the estimation method and the sample period for estimation from the "**Estimation settings**" section. Using **"@all"** instructs EViews to use the maximum sample size.

The estimation dialog box has an options tab that allows you to change the default settings for the current estimation method. Perhaps the most notable of these is the ability to control how the standard errors of the coefficients are calculated



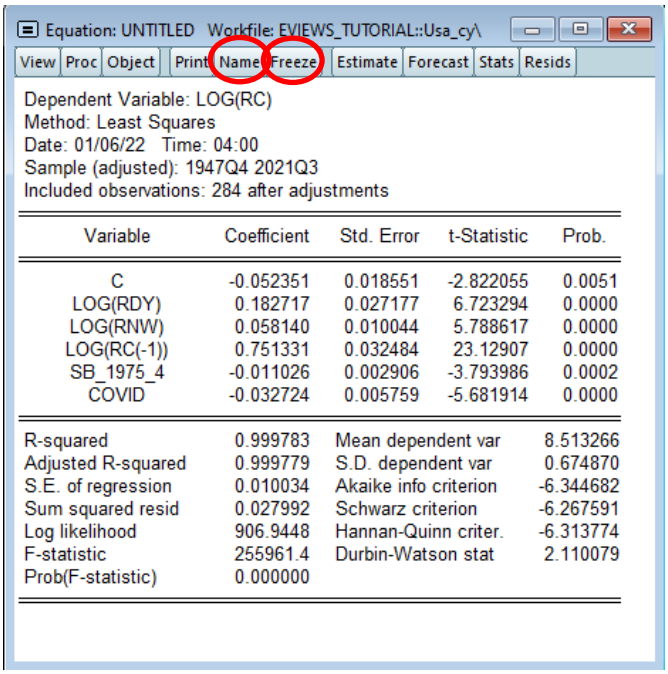
Selecting "HAC (Newey-West)" yields a hetroskedastic-autocorrelation consistent estimate of the covariance matrix of the parameter estimates without any change in the coefficient estimates themselves:

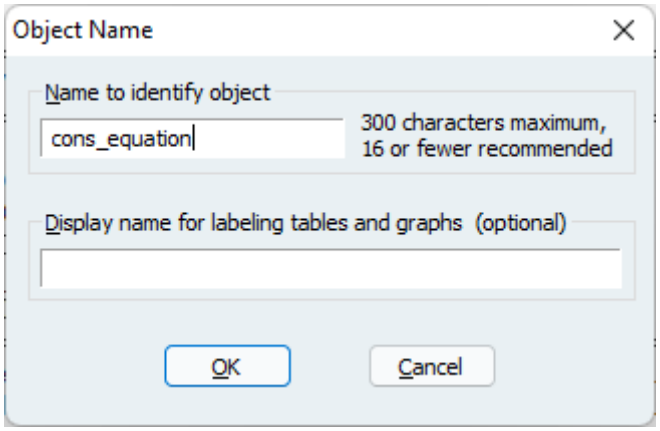


3. EQUATION OBJECTS: SAVING, LABELLING, FREEZING, PRINTING

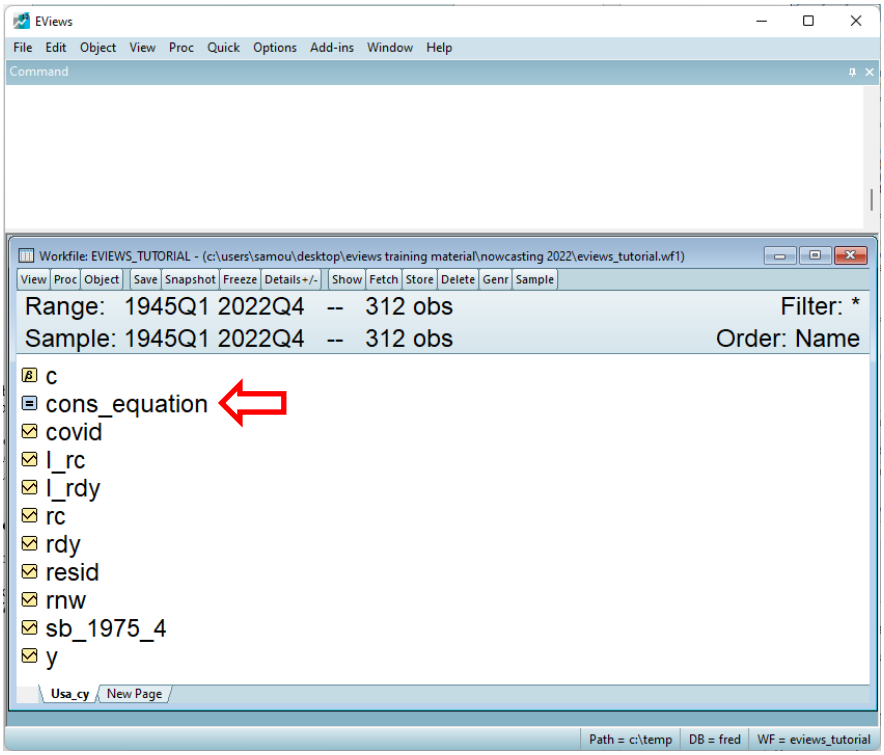
After you estimate an equation, you can save the regression/output for later retrieval.

To do so, on the Equation box, click the "**Name**" button on the top menu. The ***Object Name*** dialog box opens.



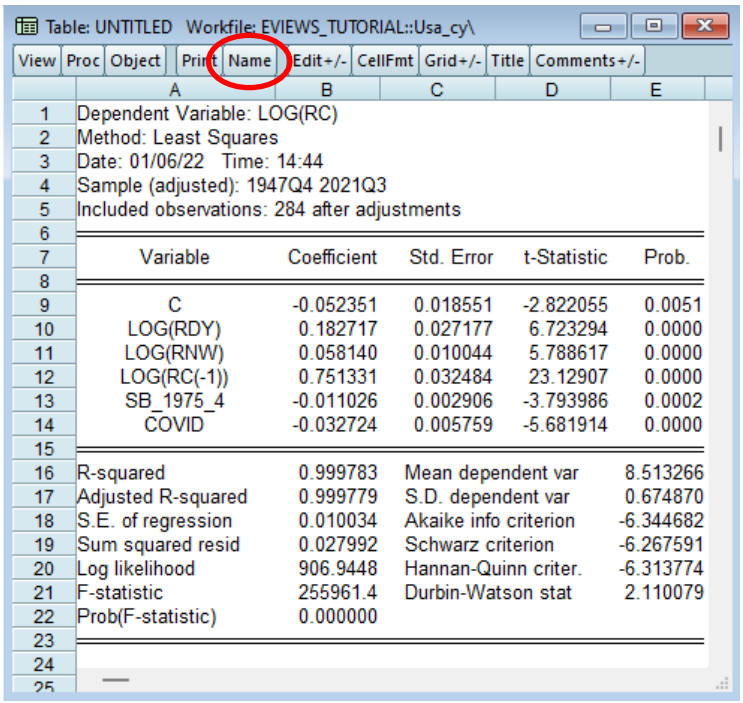


Type the name of the equation (in this case **cons_equation**) and click ***OK***. An equation object will appear in the workfile window marked with the  icon.



If you want to save the equation output so that it won't ever change (even if you re-estimate the regression), you can **Freeze** the results. Freezing the equation makes a copy of the current view in the form of a table which is detached from the Equation Object.

To freeze an object, click on the "Freeze" button on the top menu of the equation box (highlighted using a red oval in the above image). A table view of the equation results opens (next image).



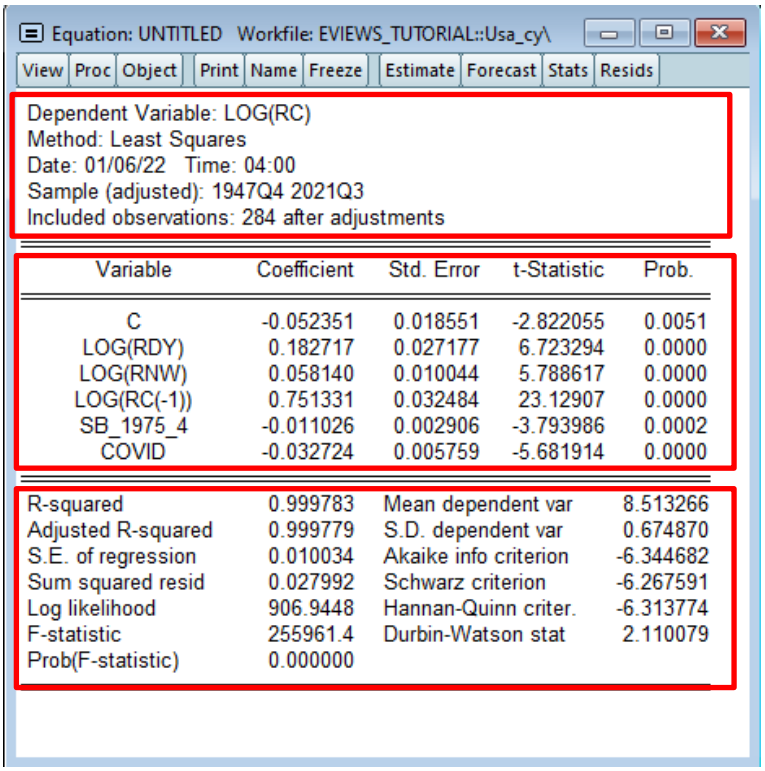
You can save this table, by clicking the "Name" button in the *Table Object* and name the table in the Object Name box (shown below; in this case we name it *table_eq1*).

4. INTERPRETING THE REGRESSION RESULTS

1 Top Panel

2 Middle Panel

3 Bottom Panel



Equation: UNTITLED Workfile: EVIEWS_TUTORIAL::Usa_cy\

View | Proc | Object | Print | Name | Freeze | Estimate | Forecast | Stats | Resids

Dependent Variable: LOG(RC)
Method: Least Squares
Date: 01/06/22 Time: 04:00
Sample (adjusted): 1947Q4 2021Q3
Included observations: 284 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	-0.052351	0.018551	-2.822055	0.0051
LOG(RDY)	0.182717	0.027177	6.723294	0.0000
LOG(RNW)	0.058140	0.010044	5.788617	0.0000
LOG(RC(-1))	0.751331	0.032484	23.12907	0.0000
SB_1975_4	-0.011026	0.002906	-3.793986	0.0002
COVID	-0.032724	0.005759	-5.681914	0.0000

R-squared	0.999783	Mean dependent var	8.513266
Adjusted R-squared	0.999779	S.D. dependent var	0.674870
S.E. of regression	0.010034	Akaike info criterion	-6.344682
Sum squared resid	0.027992	Schwarz criterion	-6.267591
Log likelihood	906.9448	Hannan-Quinn criter.	-6.313774
F-statistic	255961.4	Durbin-Watson stat	2.110079
Prob(F-statistic)	0.000000		

1) Top Panel

The **top panel** summarizes the input for the regression, including the estimation method, the effective sample, and the number of observations included in the regression.

Element	Description
Dependent Variable	Denotes the dependent variables.
Method	Denotes the method of estimation (least squares in this example).
Date/Time	Shows the date and time when the regression was carried out.
Sample	Shows the sample period over which the regression is carried out.
Included Observation	Shows the number of observations included in estimation.

2) Middle Panel

The **middle panel** summarizes the information about the regression coefficients (e.g., estimated coefficients and their standard errors).

Element	Description
Coefficient Values	• The **Log(rdy)** coefficient measures the marginal contribution of income to consumption in the initial quarter. • **C** is the estimated constant (or intercept) of the regression.
Standard Errors	• Reports the standard errors of the coefficient estimates. • The larger the standard errors, the noisier the estimates.
t-Statistic	• Reports the t-statistics, computed by dividing coefficient estimates by their standard errors. • Is used to test whether the coefficient in that row equals zero.
Prob. (p-value)	• Reports probability of drawing a t-statistic as extreme as the one estimated. • Is used to test whether the coefficient is equal to zero (against a two-sided alternative).

3) Bottom Panel

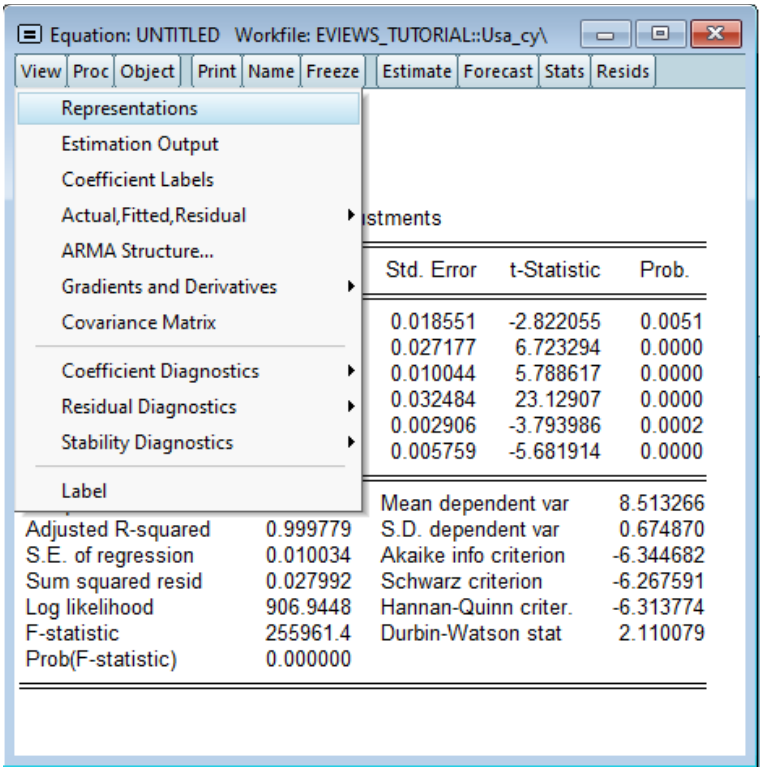
The **bottom panel** provides summary statistics about the entire regression

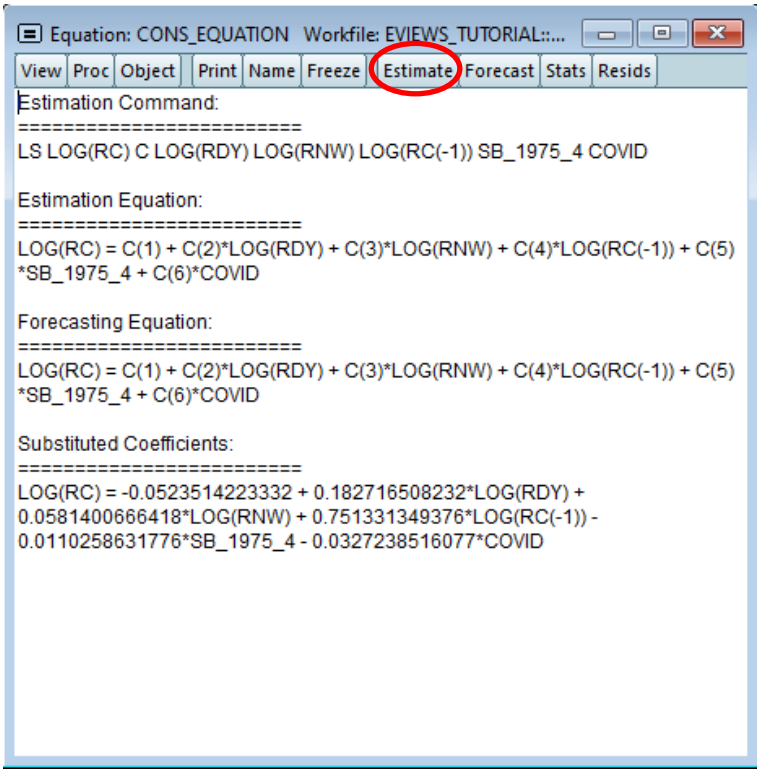
Statistic	Description
R-squared	Measures the success of the regression in predicting the values of dependent variable.
Adjusted R-squared	Adjusts for the number of independent regressors by penalizing R-squared for additional regressors.
S.E. of regression	Is a summary measure of the standard error of the estimated residuals of the regression.
Sum squared resid	Reports the sum of squared residuals. The same as $(S.E.\ of\ regression)^2 * (T-k-1)$, where T is the number of observations (284 here), k is the number of independent variables (k=6 here).
Log-likelihood	Reports the log likelihood function evaluated at coefficient estimates assuming normally distributed errors.
F-statistic	Tests whether all slope coefficients (excluding the constant) are zero.
Prob(F-Static)	Reports the probability of drawing an F-statistics as the one estimated.
Mean dependent var	Shows the mean of the dependent variable (in this case, real consumption).

S.D. dependent var	Shows the standard deviation of the dependent variable (i.e, real consumption).
Akaike info criterion	Used in model selection; smaller values are preferred. AIC is a better statistic to choose forecasting model, because it allows larger degree of freedom.
Schwarz criterion	An alternative to Akaike information (AIC) used also for model selection. Imposes a larger penalty for including additional explanatory variables
Hannan-Quinn criteria.	An alternative to AIC and Schwarz criteria used for model selection. It employs a slightly different penalty function than the other two.
Durbin-Watson stat	Measures serial correlation in the residuals. As a rule of thumb, a DW statistic less than 2 is an indication of positive serial correlation.

5. VIEW → REPRESENTATION

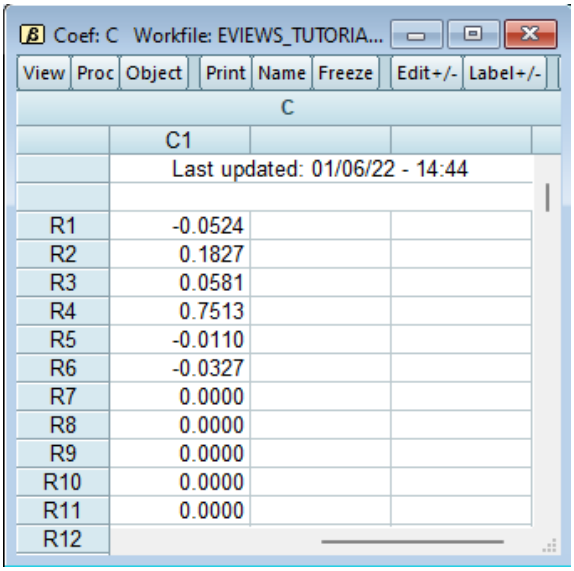
This command lists the command line, the algebraic representation, and the estimated regression line of the most recently estimated regression. It is invoked from the regression output box:





Notice that the estimated parameters of the regression are automatically given the names **C(1)** to **C(6)**. This means that they are the first six elements of the "**c**" vector in the **USA_cy** pagefile.

Double click on the "c" series to confirm this. You should see the following screen:

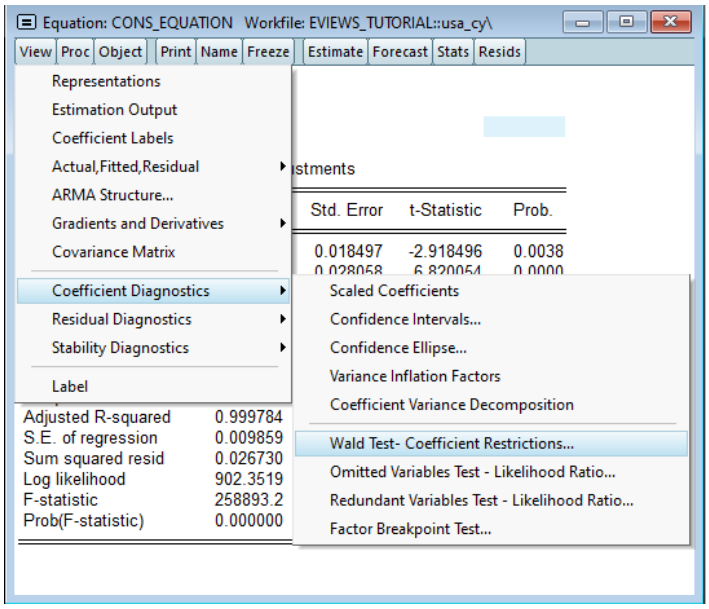


Lastly, clicking on the **"Estimate"** tab of the equation representation output box brings you back to the equation box where you can change the regression **Options,** such as the way the standard errors of the coefficient estimates are calculated (see below).

6. HYPOTHESIS TESTING

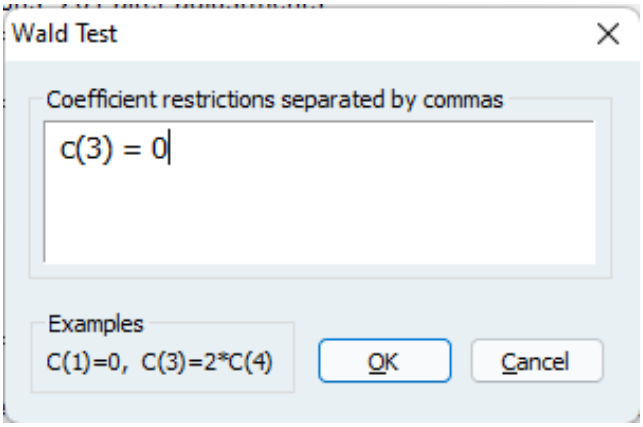
EViews allows you to test a complex hypothesis on the parameter space very easily. To accomplish this, click

View → Coefficient Diagnostics → Wald Test → Coefficient Restrictions



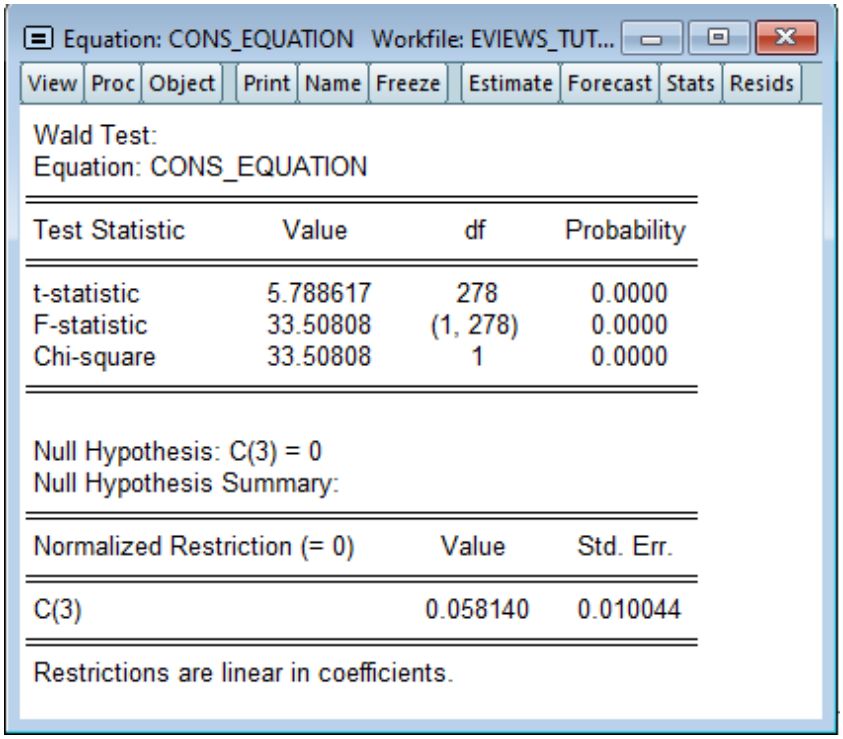
The Wald Test dialog box opens. As we saw previously, EViews names the estimated coefficients c(1), c(2), c(3), etc., numbering them in the order they appear in the regression (including the constant). For example, in the consumption equation, the coefficient on **log(rnw)** is c(3).

To test whether the true coefficient is zero involves testing the null hypothesis that c(3) = 0. You can specify this hypothesis as an equation in the Wald Test box by typing **c(3)=0.**



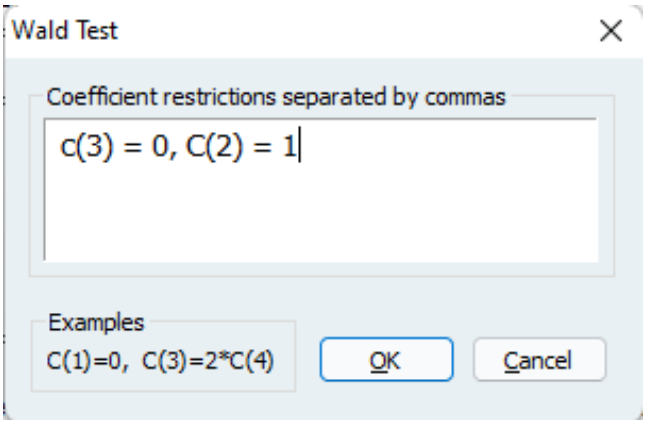
EViews tests this hypothesis using the Wald statistic, which can be shown to have an F and, equivalently, a Chi-Square distribution under the null hypothesis that the restrictions are valid.

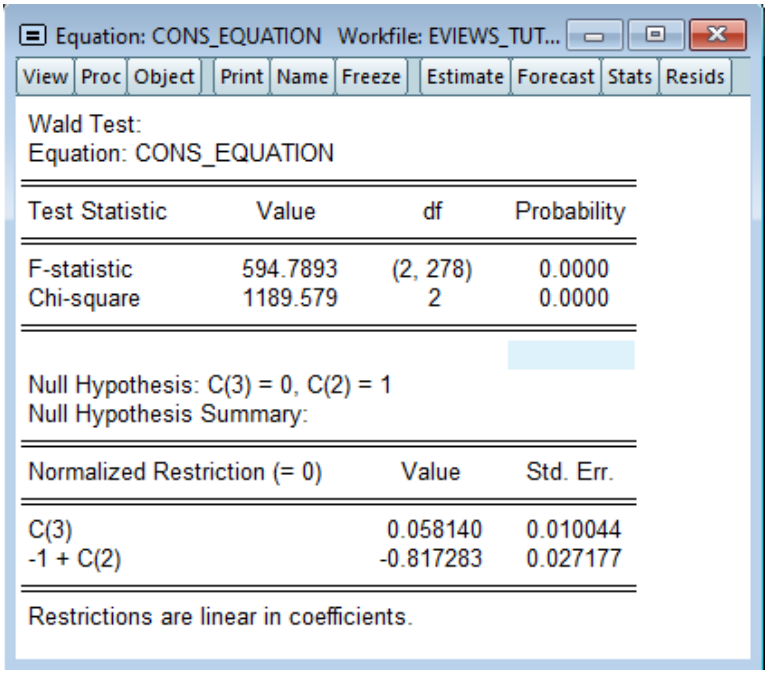
The result is:



Note that for single (linear restrictions), the **t-statistic** in the output page is identical to the t-statistic shown in the regression output in the "t-statistic" column. Also, the F and Chi-Square statistics are the square of the t-statistic.

For multiple restrictions, enter multiple coefficient equations, separated by commas. For example, in addition to testing whether real wealth has an impact on real consumption, you can also test whether the marginal propensity to consume (i.e., the coefficient on real disposable income, or c(2) in this case) is also equal to one.



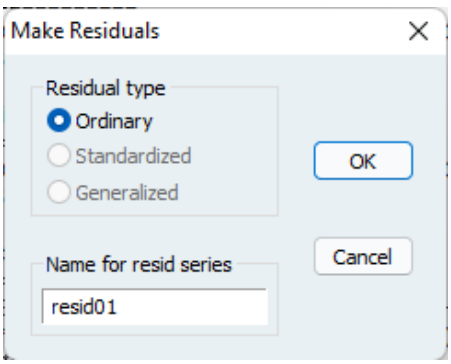


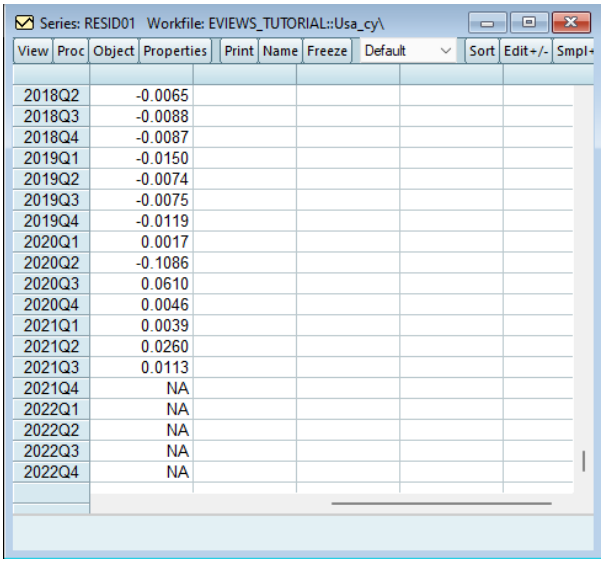
Note that the t-statistic is no longer presented. This is because the t-statistic is meant for hypothesis tests involving a single hypothesis.

7. USING THE REGRESSION WINDOW

1) Make Residual Series

You can easily create a residual series for the **Forecasts (fitted values)** for the current/active sample using **Proc → Make Residual Series…** Simply enter a name for the residual series in the subsequent dialog box (default is "resid01", but you should change this to something that is more familiar to you):

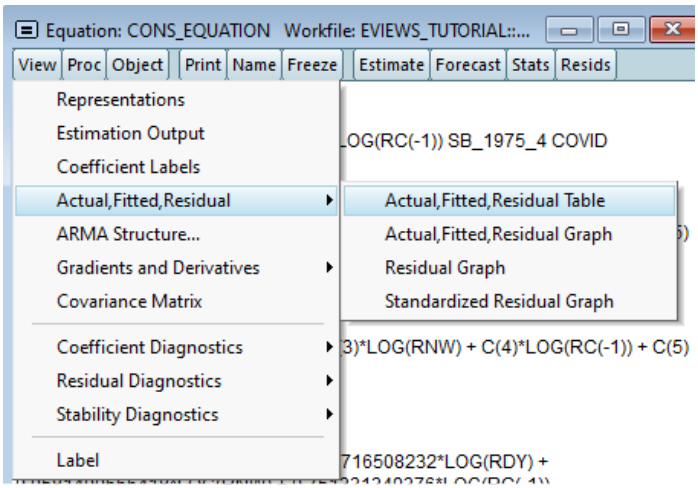


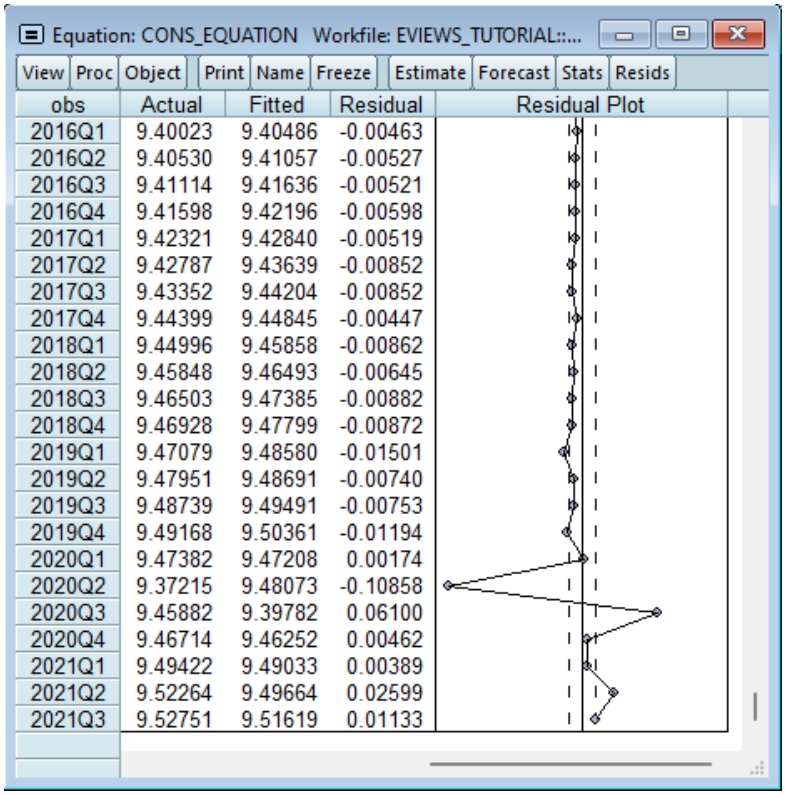


2) Show Actual, Fitted, Residual

You can obtain the actual values of the dependent together with the fitted values and residuals using

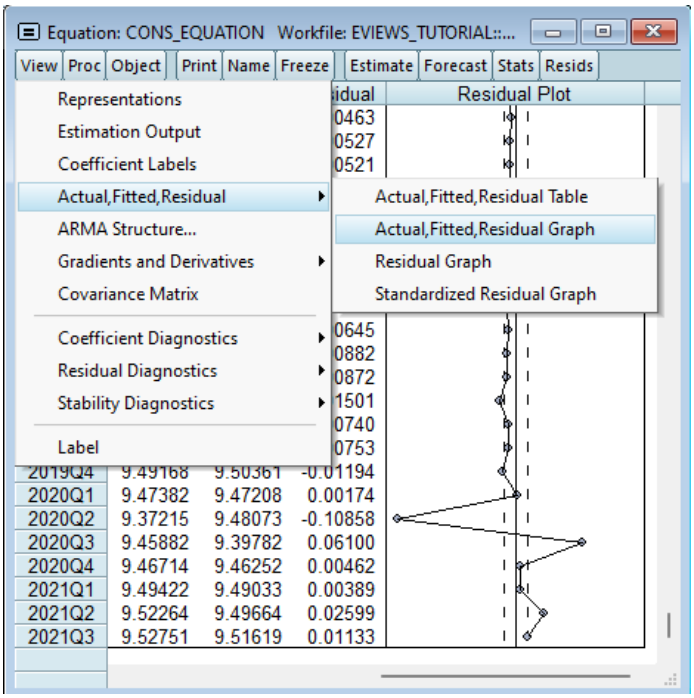
View → Actual, Fitted, Residual → Actual, Fitted, Residual Table

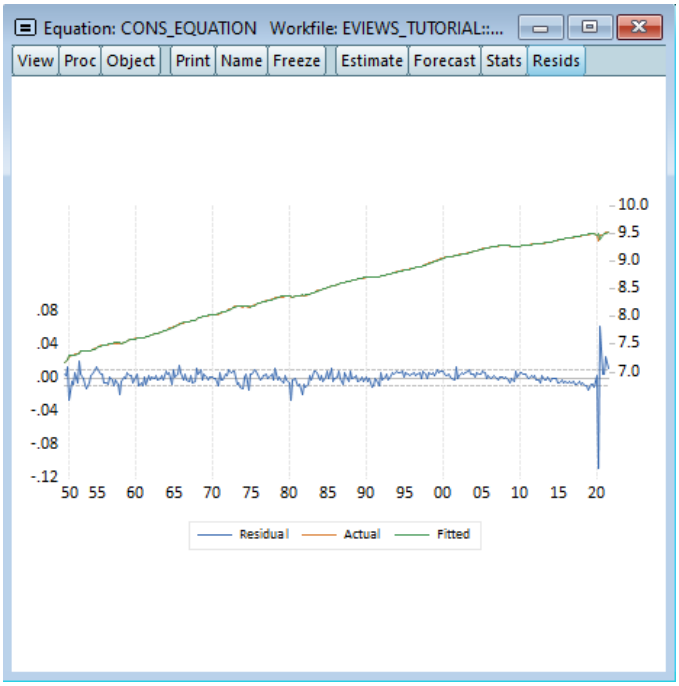




Alternatively, if you prefer a visual representation:

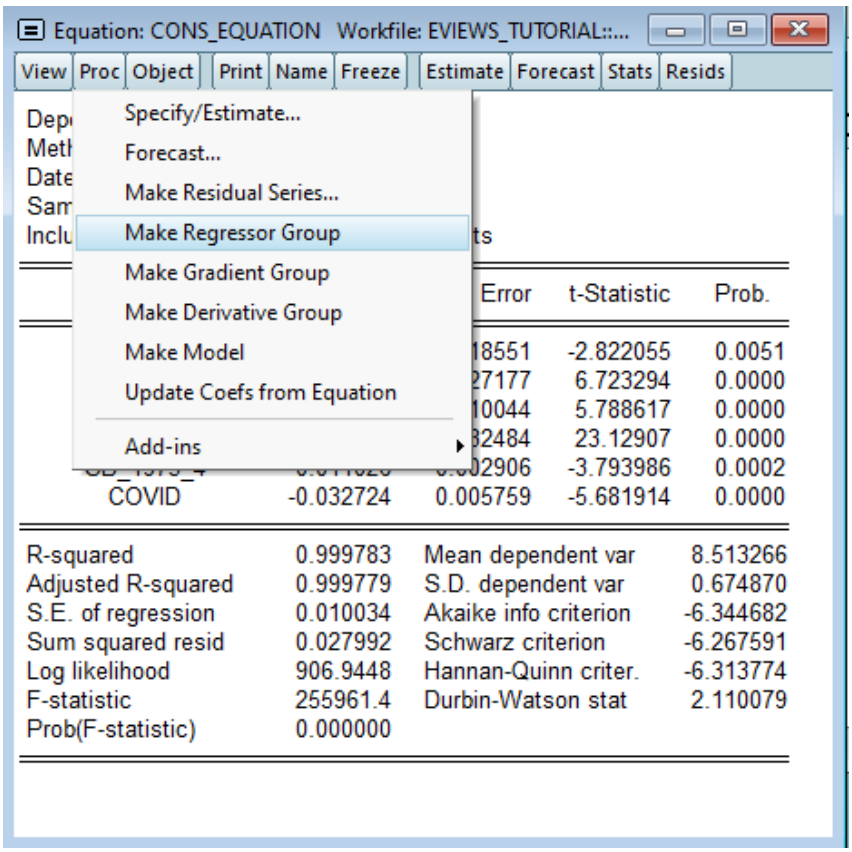
View → Actual, Fitted, Residual → Actual, Fitted, Residual Graph

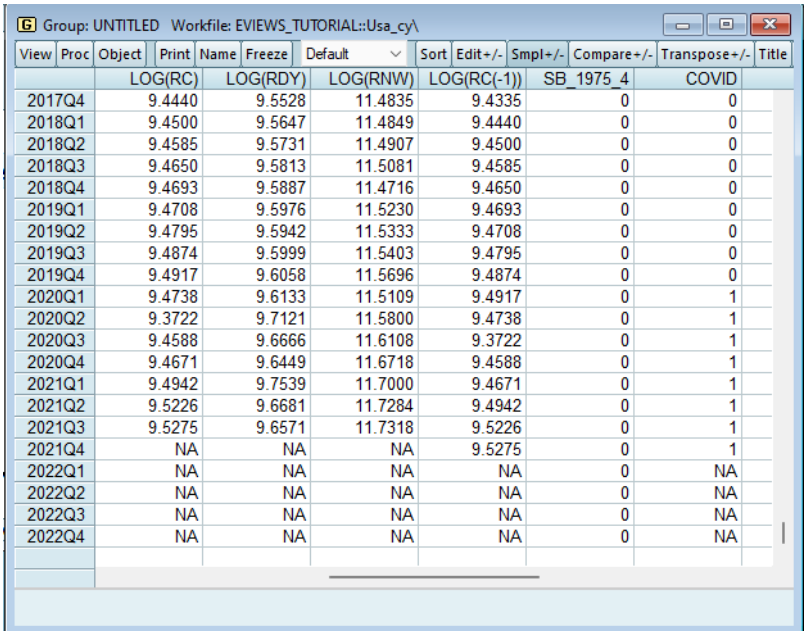




3) Make Regressor Group

The "Proc" menu has a useful feature that allows you to inspect the data used to estimate the regression. Click on **Proc → Make Regressor Group**





Group: UNTITLED Workfile: EVIEWS_TUTORIAL::Usa_cy\

View | Proc | Object | Print | Name | Freeze | Default | Sort | Edit+/- | Smpl+/- | Compare+/- | Transpose+/- | Title

	LOG(RC)	LOG(RDY)	LOG(RNW)	LOG(RC(-1))	SB_1975_4	COVID
2017Q4	9.4440	9.5528	11.4835	9.4335	0	0
2018Q1	9.4500	9.5647	11.4849	9.4440	0	0
2018Q2	9.4585	9.5731	11.4907	9.4500	0	0
2018Q3	9.4650	9.5813	11.5081	9.4585	0	0
2018Q4	9.4693	9.5887	11.4716	9.4650	0	0
2019Q1	9.4708	9.5976	11.5230	9.4693	0	0
2019Q2	9.4795	9.5942	11.5333	9.4708	0	0
2019Q3	9.4874	9.5999	11.5403	9.4795	0	0
2019Q4	9.4917	9.6058	11.5696	9.4874	0	0
2020Q1	9.4738	9.6133	11.5109	9.4917	0	1
2020Q2	9.3722	9.7121	11.5800	9.4738	0	1
2020Q3	9.4588	9.6666	11.6108	9.3722	0	1
2020Q4	9.4671	9.6449	11.6718	9.4588	0	1
2021Q1	9.4942	9.7539	11.7000	9.4671	0	1
2021Q2	9.5226	9.6681	11.7284	9.4942	0	1
2021Q3	9.5275	9.6571	11.7318	9.5226	0	1
2021Q4	NA	NA	NA	9.5275	0	1
2022Q1	NA	NA	NA	NA	0	NA
2022Q2	NA	NA	NA	NA	0	NA
2022Q3	NA	NA	NA	NA	0	NA
2022Q4	NA	NA	NA	NA	0	NA

This view can be especially helpful when the results from the regression are not consistent with priors. Inspecting and (especially) graphing the data very quickly reveals errors in construction or data alignment.

4) Saving your results

Each time you generate a table or a graph, you can save the result in your workfile using **Freeze** then **Name** in the new window. Note that these views are not saved by default. They will need to be re-created if you do not save them to the workfile before exiting EViews.

8. TESTING FOR SERIAL CORRELATION

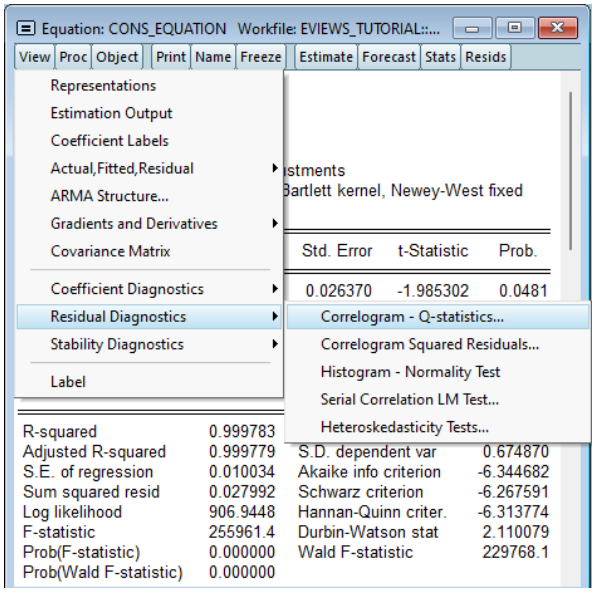
Serial correlation is a common occurrence in time series data because the data is ordered (over time). It is therefore not surprising that neighboring error terms turn out to be correlated. Serial correlation violates the standard assumption of regression theory that error terms are uncorrelated.

If untreated, serial correlation leads to several issues, including reported standard errors and t-statistics are invalid (even asymptotically), and coefficient estimates that may be biased, though not necessarily inconsistent (assuming the data is weakly dependent). In the presence of lagged dependent variables, OLS estimates are biased and inconsistent, making it essential to test whether there is serial correlation in the residuals.

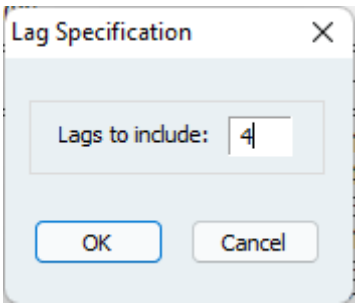
EViews provides several tools for detecting serial correlation and correcting regressions to account for its presence.

1) Correlogram Q Statistics

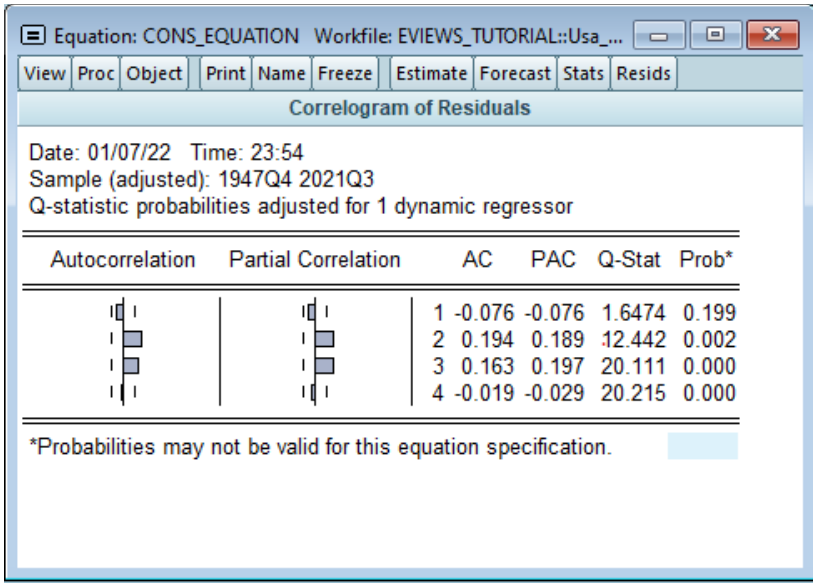
The first is the correlogram Q statistic of the residuals, calculated from the equation box using "**View → Residual Diagnostics → Correlogram Q Statistics…**":



The lag selection box opens to allow you to specify the number of lags of the residuals to include in the calculation of the Q-Statistic. We shall use 4 lags (by assumption).



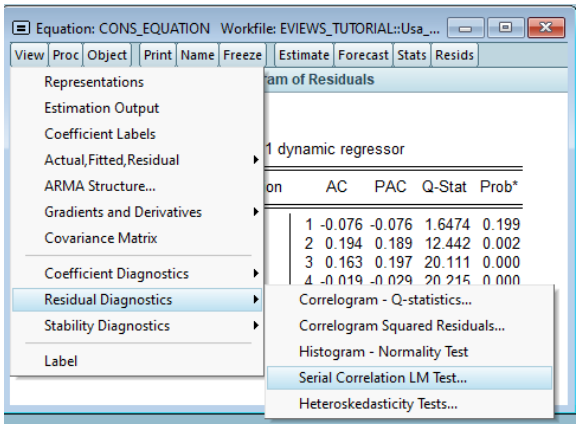
The resulting output screen is



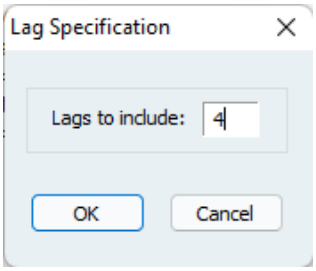
The Q statistic rejects the null hypothesis of no-autocorrelation in the residuals. Notice that the 2nd, 3rd, and 4th lags are all statistically significant, as is evident from the low p-value of the statistic (see "Prob*" column).

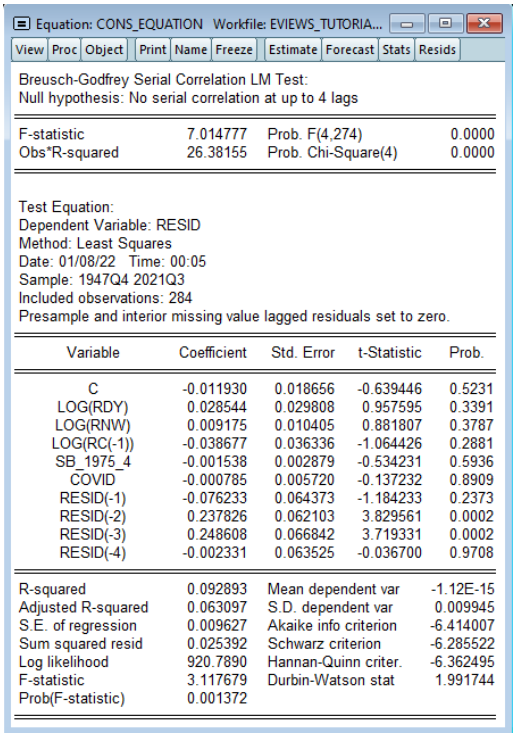
2) Bruesch-Godfrey LM Test

A more general test for serial correlation is the Bruesch-Godfrey LM test, which is available in EViews: **View→Residual Diagnostics→ Serial Correlation LM test**



As before, the lag selection box opens, and you need to specify the number of lags to use for the Bruesch-Godfrey test (4 in this example):





Equation: CONS_EQUATION Workfile: EVIEWS_TUTORIA...

View | Proc | Object | Print | Name | Freeze | Estimate | Forecast | Stats | Resids

Breusch-Godfrey Serial Correlation LM Test:
Null hypothesis: No serial correlation at up to 4 lags

F-statistic	7.014777	Prob. F(4,274)	0.0000
Obs*R-squared	26.38155	Prob. Chi-Square(4)	0.0000

Test Equation:
Dependent Variable: RESID
Method: Least Squares
Date: 01/08/22 Time: 00:05
Sample: 1947Q4 2021Q3
Included observations: 284
Presample and interior missing value lagged residuals set to zero.

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	-0.011930	0.018656	-0.639446	0.5231
LOG(RDY)	0.028544	0.029808	0.957595	0.3391
LOG(RNW)	0.009175	0.010405	0.881807	0.3787
LOG(RC(-1))	-0.038677	0.036336	-1.064426	0.2881
SB_1975_4	-0.001538	0.002879	-0.534231	0.5936
COVID	-0.000785	0.005720	-0.137232	0.8909
RESID(-1)	-0.076233	0.064373	-1.184233	0.2373
RESID(-2)	0.237826	0.062103	3.829561	0.0002
RESID(-3)	0.248608	0.066842	3.719331	0.0002
RESID(-4)	-0.002331	0.063525	-0.036700	0.9708

R-squared	0.092893	Mean dependent var	-1.12E-15
Adjusted R-squared	0.063097	S.D. dependent var	0.009945
S.E. of regression	0.009627	Akaike info criterion	-6.414007
Sum squared resid	0.025392	Schwarz criterion	-6.285522
Log likelihood	920.7890	Hannan-Quinn criter.	-6.362495
F-statistic	3.117679	Durbin-Watson stat	1.991744
Prob(F-statistic)	0.001372		

The LM statistic confirms the presence of autocorrelation in the residuals. The F-statistic. With 4 lags, the statistic has F-statistic of 7.01 and a p-value of 0.00. Hence, we reject the null hypothesis of no-autocorrelation.

Serial correlation in the error term may be evidence of a serious problem of model misspecification. If your goal is to estimate a model with complete dynamics, you need to re-specify the model. If you do not wish to estimate a fully dynamic model, but would like to carry out statistical inference, then you need to account for serial correlation so that test statistics are valid. Note that EViews has several built-in procedures to correction for serial correlation in the residuals, including the addition of AR or MA terms in the regression to be estimated.

9. OUT-OF-SAMPLE FORECASTING EXPERIMENT-BASELINE FORECAST

EViews offers a powerful and easy-to-use forecasting tool that allows you to obtain forecasts from your estimated models. Of course, the accuracy of the forecasts depends on the model used to produce the forecasts: EViews simply handles the mechanics of producing the forecasts.

1) Forecast using "Forecast" function

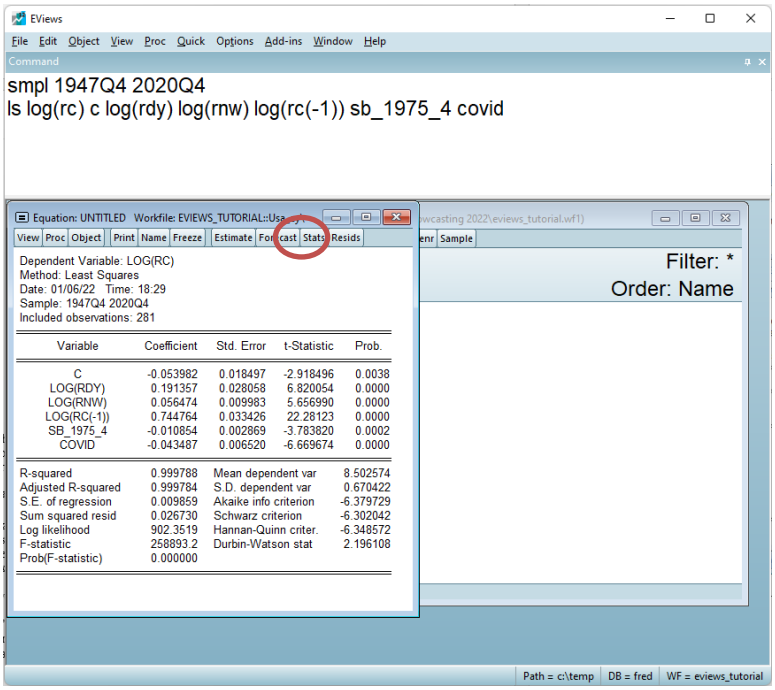
It's relatively easy to conduct out-of-sample forecast experiments using EViews. The **first step** is to set the active sample to something slightly shorter than the available data, viz (for the **USA_CY** data), say:

smpl 1947Q1 2020Q4

leaving three known observations outside of estimation sample (i.e., those for 2020Q1-2021Q3) to evaluate the regressions forecasting performance.

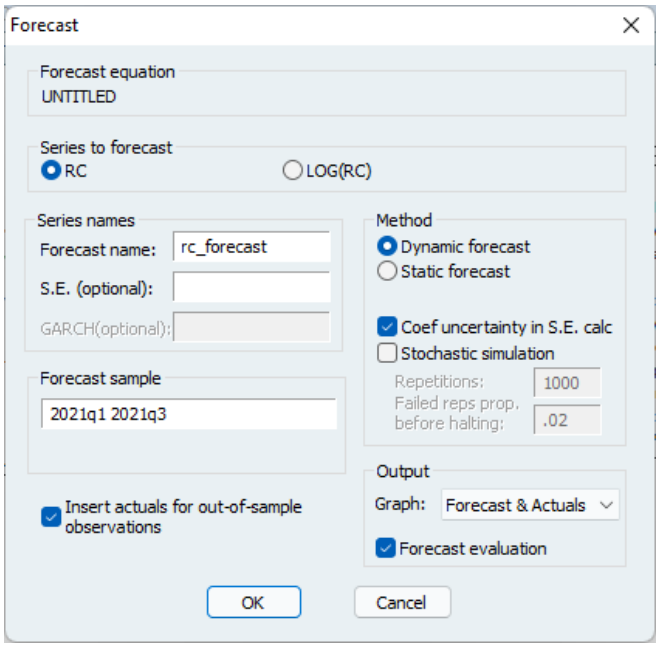
The **second step** is to forecast the equation over this sub-sample using OLS.

ls log(rc) c log(rdy) log(rnw) log(rc(-1)) sb_1975_4 covid

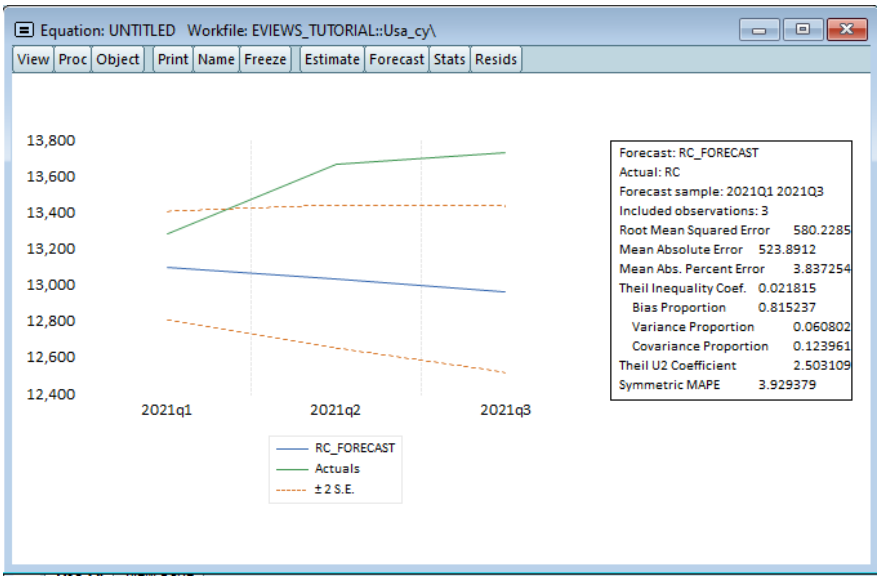


The **third step** is to generate the forecast. If you are using a single equation, you can generate the forecast in two ways:

(a) Click on the "**Forecast**" button (see above) to invoke the forecast dialog box:



Notice that, for forecasting purposes, the sample is 2021Q1-2021Q3, which is outside of the current estimation period. Also, because the model has a lagged dependent variable, namely **log(rc(-1))**, EViews allows you to choose between dynamic and static forecasting. Static forecasting uses the actual value of the lagged dependent variable (if it is available), while dynamic forecasts use the forecasted value of the lagged dependent variable, hence allowing the forecast errors to accumulate the longer the forecast horizon is. For out of sample forecasting, dynamic forecasting is usually the only possible approach (due to the lack of actual data, static forecasting is impossible).



Suppose the forecast sample is $j = T+1, T+2, \ldots, T+h$, and denote the actual and forecasted value in period t as y_t and $\hat{y}_t$, respectively. The reported forecast error statistics are computed as follows:

Root Mean Squared Error	$\sqrt{\sum_{t=T+1}^{T+h} (\hat{y}_t - y_t)^2 / h}$
Mean Absolute Error	$\sum_{t=T+1}^{T+h} \lvert \hat{y}_t - y_t \rvert / h$
Mean Absolute Percentage Error	$100 \sum_{t=T+1}^{T+h} \left\lvert \frac{\hat{y}_t - y_t}{y_t} \right\rvert / h$
Theil Inequality Coefficient	$\frac{\sqrt{\sum_{t=T+1}^{T+h} (\hat{y}_t - y_t)^2 / h}}{\sqrt{\sum_{t=T+1}^{T+h} \hat{y}_t^2 / h} + \sqrt{\sum_{t=T+1}^{T+h} y_t^2 / h}}$
Theil U2 Statistics	Theil U2 statistics measure forecast quality relative to the naïve model. A Theil U2 equal to 1 indicates a

	forecasting performance equivalent to that of the naïve forecasting approach (random walk). It is preferred to the U1 statistic because the U1 statistic cannot exceed 1.0. A smaller value than 1.0 implies accuracy superior to the naïve model, whereas a value greater than 1 indicates a less accurate model compared to the naïve model.

The first two forecast error statistics depend on the scale of the dependent variable. These statistics should be used as relative measures to compare forecasts for the same series across different models. The remaining three statistics are scale invariant.

Moreover, the forecast target is "**rc**" rather than "**log(rc)",** which is the actual dependent variable of the estimated regression.

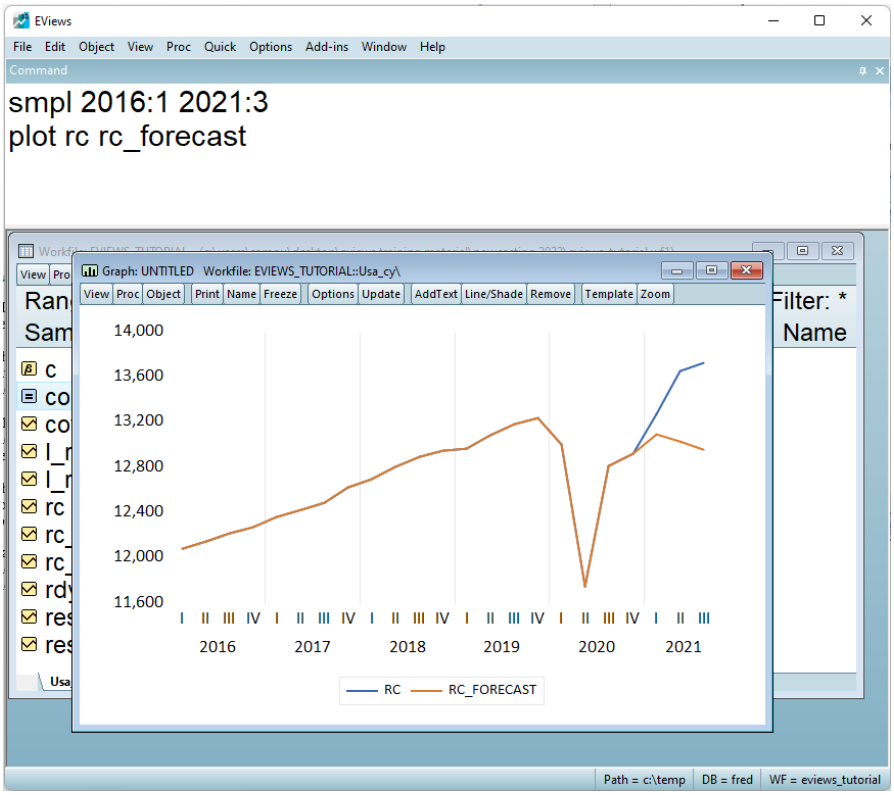
The forecasts will be saved in the workfile in a series object called "**rc_forecast**" rather than "**rc**" to avoid overwriting the original series. You can change the name of the forecast variable if you like.

Click on OK to generate the (dynamic) forecast, plot the actual values of **rc** against the forecast, and obtain a table of summary statistics that allow you to assess the performance of the forecasting model relative to another model.

The results suggest that the fitted regression is systematically under-estimating actual consumption during the 2021Q1-2021Q3 period. This is because the forecast errors for the period are all positive.

Lastly, a better perspective on the magnitude of the (dynamic) forecast error can be obtained by increasing the sample size over which actual consumption and fitted consumption are plotted. To do so, plot **rc** and **rc_forecast** for the period 2016Q1-2021Q3.

```
smpl 2016Q1 2021Q3
plot rc rc_forecast
```



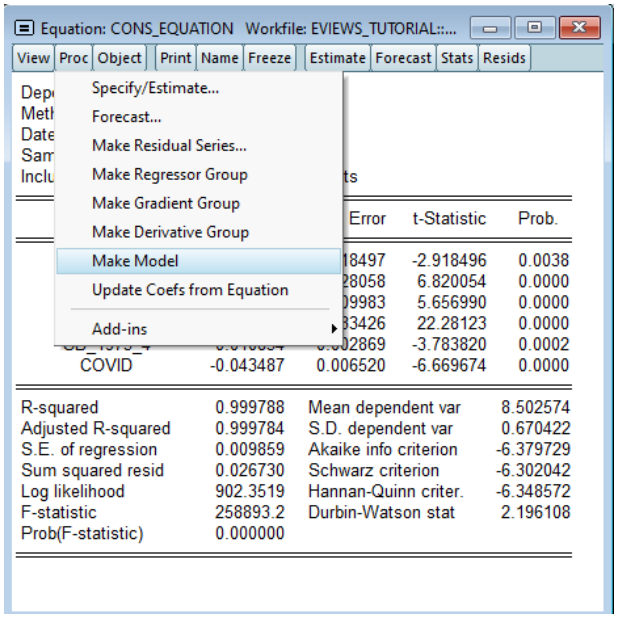
The equation is clearly underestimating real consumption significantly during 2021, suggesting the marginal propensity to consume was indeed higher than estimated during the COVID period and the need for re-specification. One option would be to re-estimate the model as a VAR and/or error-correction model using EViews VAR routines.

2) Forecast using model "Solver" routine

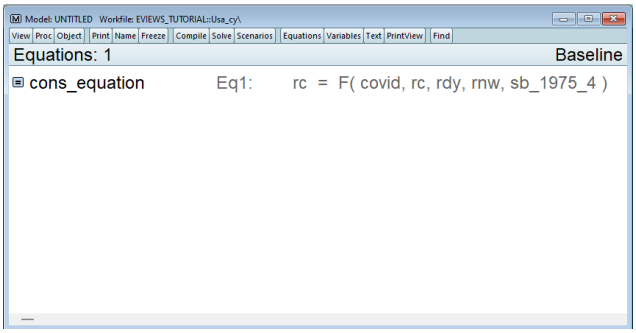
The second approach to generating the forecast is to use EViews powerful model solver routines, which we will be making good use of throughout this course. The following provides an example of how it can be used to do the same forecasting exercise as above.

To do so, re-run first the regression again over 1947Q1-2020Q4 and then invoke

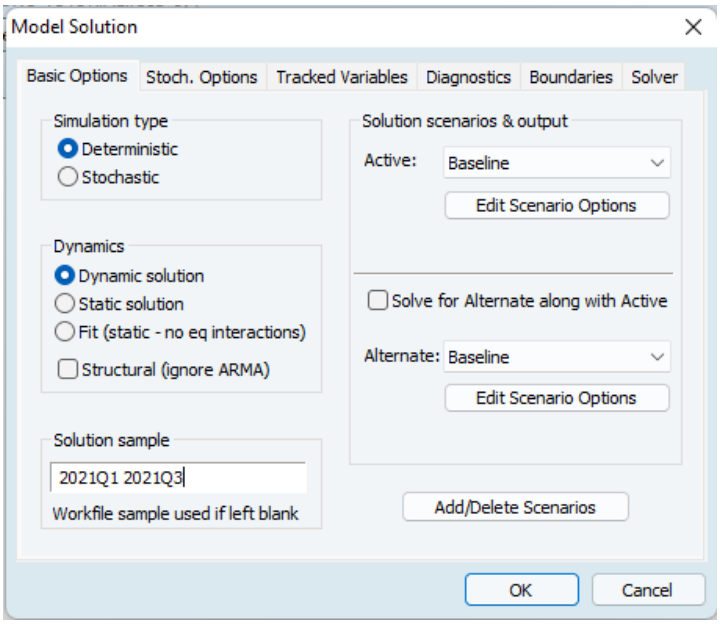
Proc → Make Model.



You should see the following output…

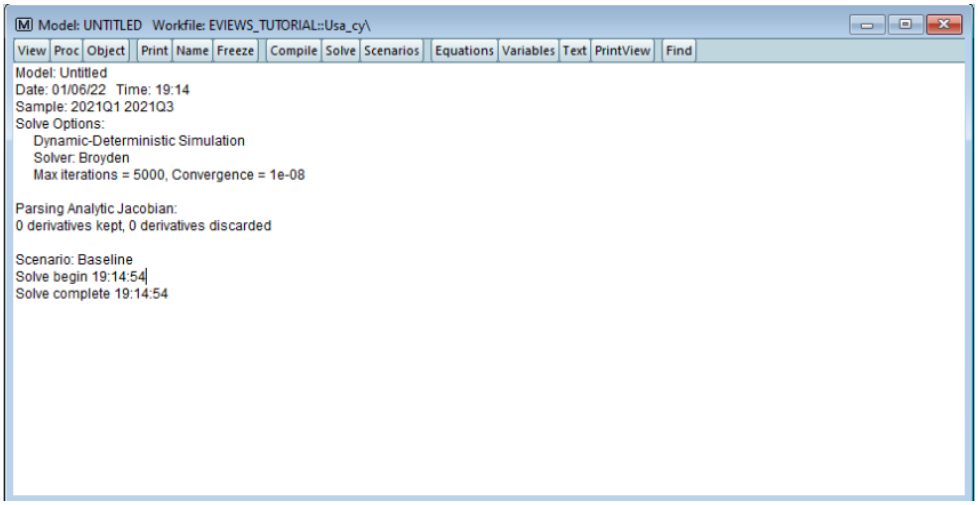


Now click on "**Solve**" to solve the model (i.e., generate the baseline solution) over the forecast period, which is 2021:1 2021:3.

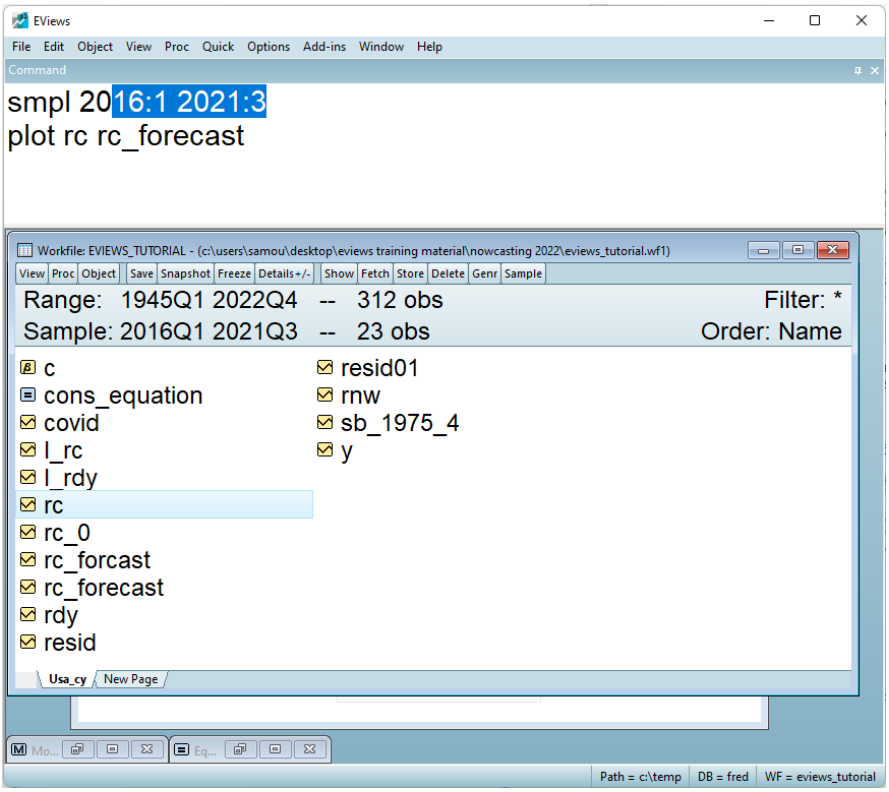


By default, EViews will save the solution (the so-called "baseline" or forecasted series) with the same name as the dependent variable (in this case "**rc**") but with **"_0"** appended to it – thus "**rc_0**". It is highly recommended that you do not to change these defaults.

Press **OK** to calculate the forecast.

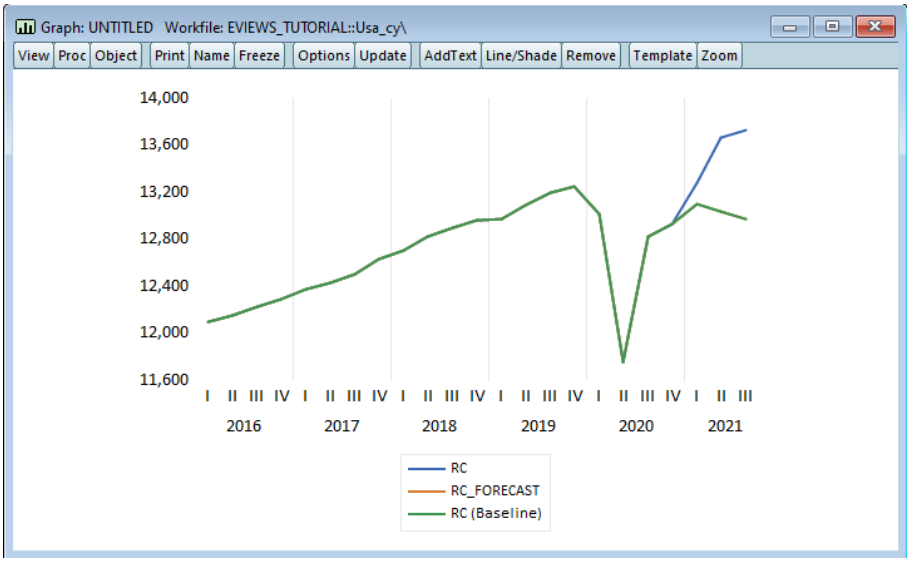


Notice that the workfile now has a series called "**rc_0**"



Now graph **rc, rc_0**, and **rc_forecast** (from the first approach) to assess whether the forecasts from the different approaches are identical.

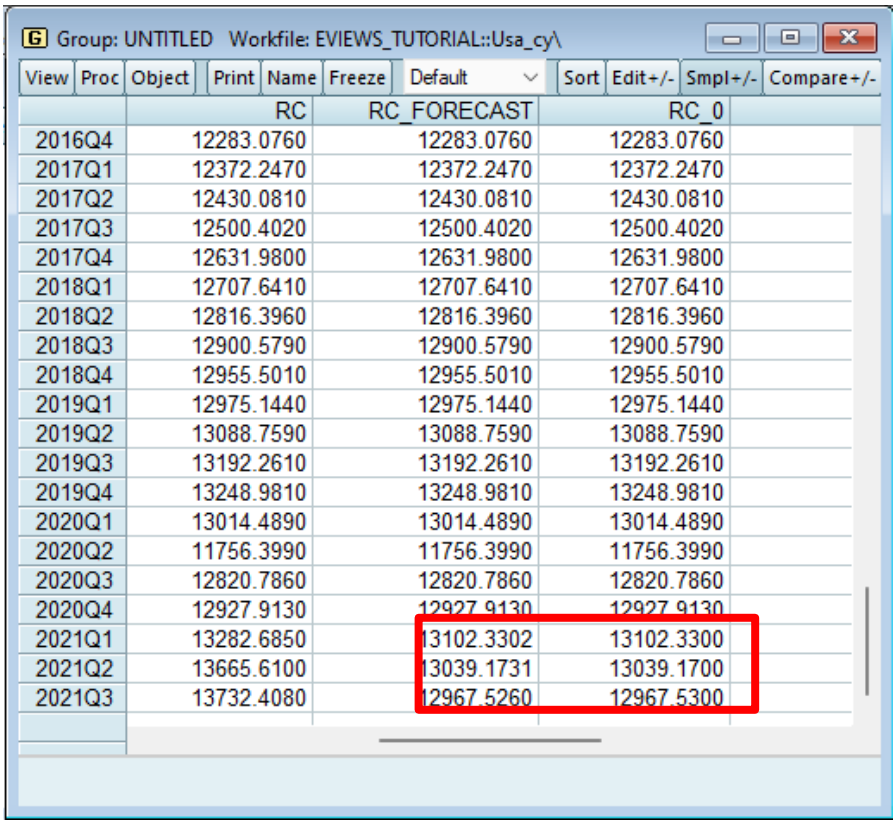
smpl 2016Q1 2021Q3
plot rc rc_forecast rc_0



As to be expected, "**rc_0"** (i.e., the baseline) cannot be distinguished from **"rc_forecast"** in the above graph. They are identical, as required.

Indeed, they have the same values, as the following view verifies:

show rc rc_forcast rc_0



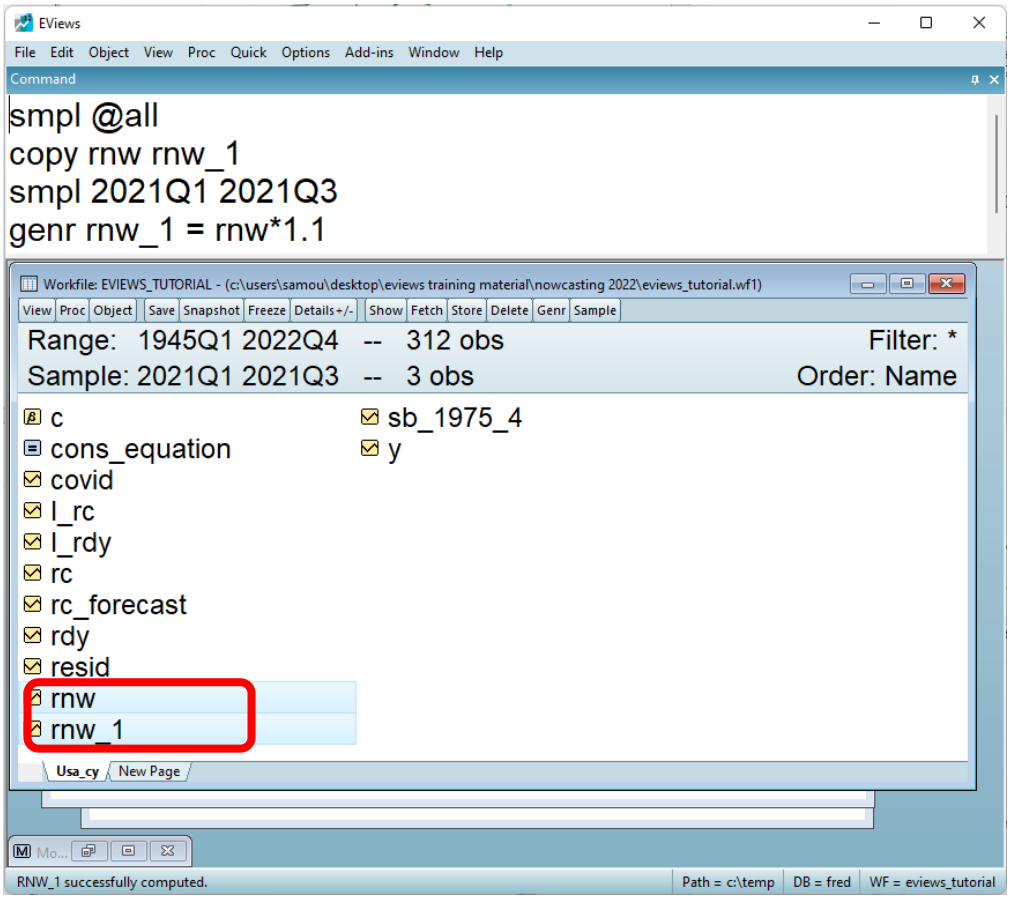
	RC	RC_FORECAST	RC_0
2016Q4	12283.0760	12283.0760	12283.0760
2017Q1	12372.2470	12372.2470	12372.2470
2017Q2	12430.0810	12430.0810	12430.0810
2017Q3	12500.4020	12500.4020	12500.4020
2017Q4	12631.9800	12631.9800	12631.9800
2018Q1	12707.6410	12707.6410	12707.6410
2018Q2	12816.3960	12816.3960	12816.3960
2018Q3	12900.5790	12900.5790	12900.5790
2018Q4	12955.5010	12955.5010	12955.5010
2019Q1	12975.1440	12975.1440	12975.1440
2019Q2	13088.7590	13088.7590	13088.7590
2019Q3	13192.2610	13192.2610	13192.2610
2019Q4	13248.9810	13248.9810	13248.9810
2020Q1	13014.4890	13014.4890	13014.4890
2020Q2	11756.3990	11756.3990	11756.3990
2020Q3	12820.7860	12820.7860	12820.7860
2020Q4	12927.9130	12927.9130	12927.9130
2021Q1	13282.6850	13102.3302	13102.3300
2021Q2	13665.6100	13039.1731	13039.1700
2021Q3	13732.4080	12967.5260	12967.5300

3) Out of Sample Experiment: Alternative Scenario (10 percent Increase in Volume)

Suppose that you want to investigate the impact of a 10 percent increase in real net worth during 2021Q1-2021Q3 on the forecast for real consumption. You can do this easily by creating a specifically named variable reflecting this shock and then instructing EViews to use this variable to compute the (alternative) forecast. Doing this is called "Creating an Alternative Scenario" typically goes by the name "Scenario #" where # is replaced by an integer (e.g., 1). The simulator can be set up to solve both the baseline and the alternative scenario simultaneously.

Calculate the variables that reflect the maintained assumptions of the alternative scenario. In our case, we will assume that real net worth, **rnw**, is higher by 10 percent during 2021Q1-2021Q3. This can be done easily using EViews' GENR command:

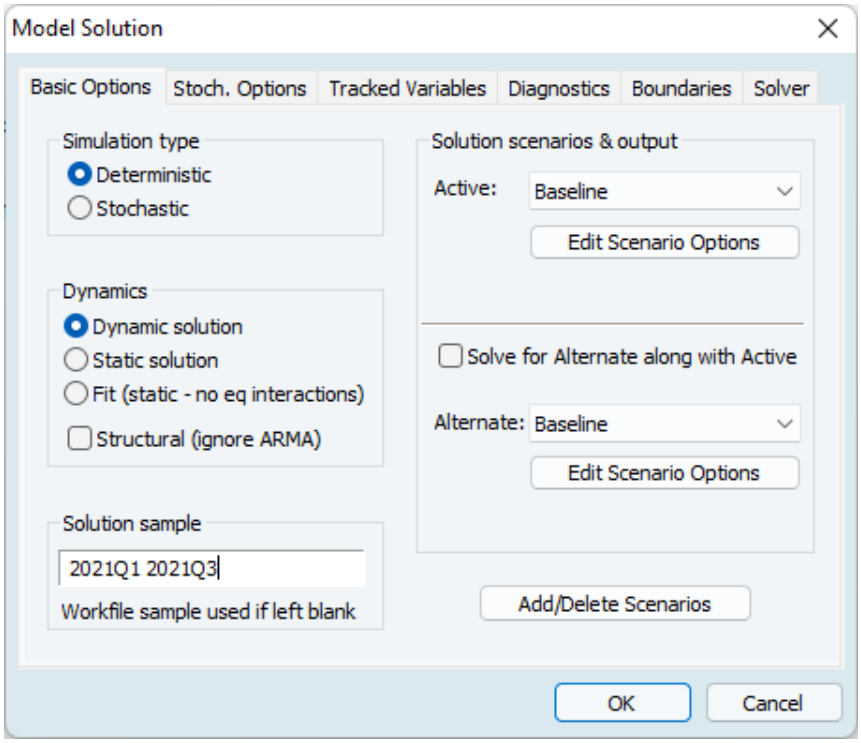
smpl @all
copy rnw rnw_1
smpl 2021Q1 2021Q3
genr rnw_1 = rnw*1.1



The fact that **rnw_1** is greater than **rnw** during 2021Q1-2021Q3 can be verified by plotting both on a single graph, namely: **plot rnw rnw_1**

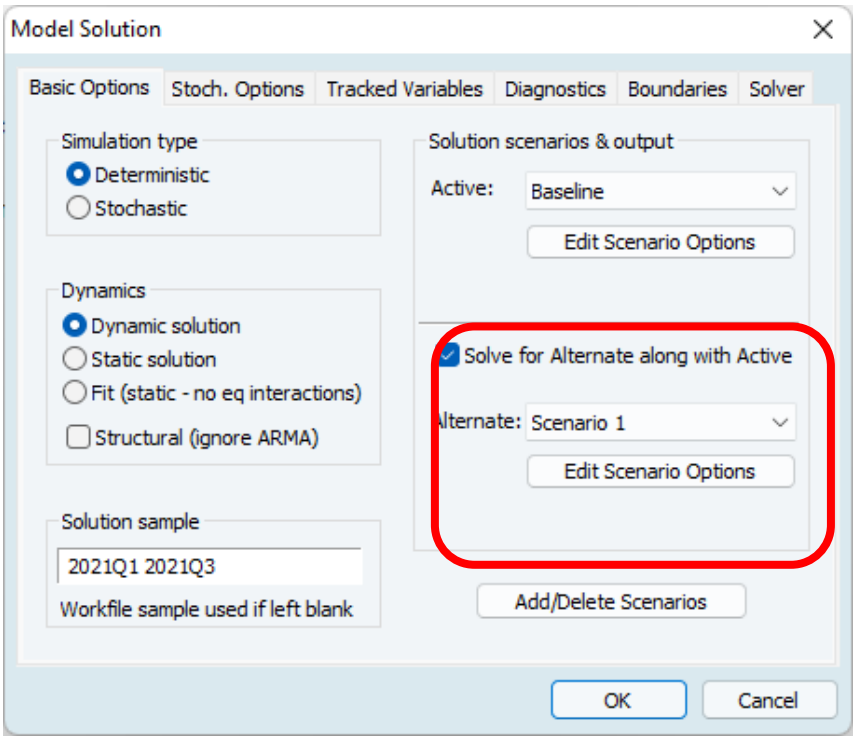


Now return to the EViews model simulator and the model object and click on **Solve**. You should again see the following dialog box:

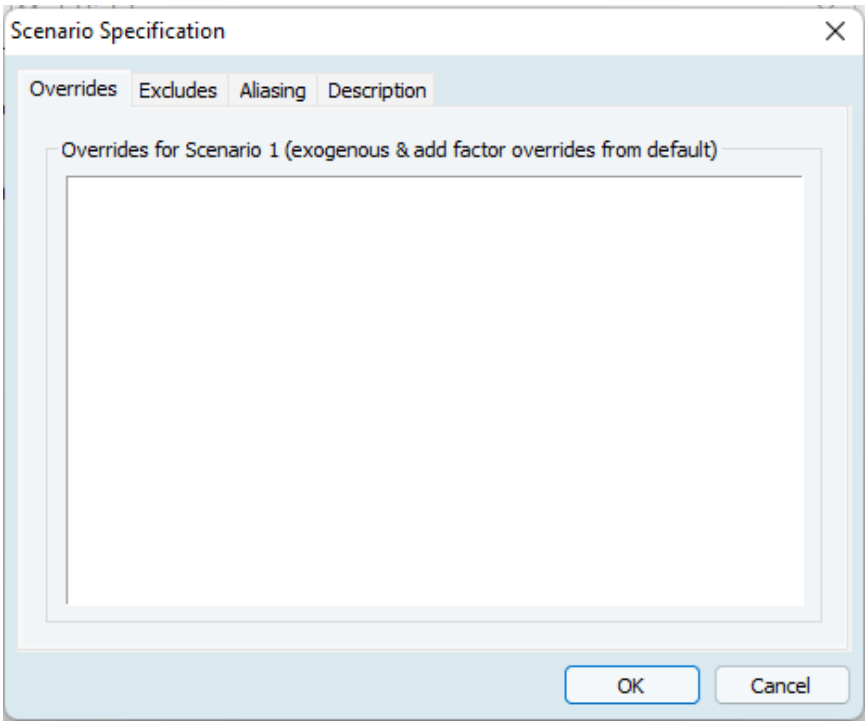


By default, EViews will always solve the baseline scenario (i.e., the active scenario) unless instructed otherwise (by, for example, making the active scenario an alternative scenario (not recommended)).

Now check the "Solve for Alternate along with Active" box to get EViews to calculate (along with the baseline) and alternative scenario and make the Alternate scenario: "Scenario 1". You should now see:

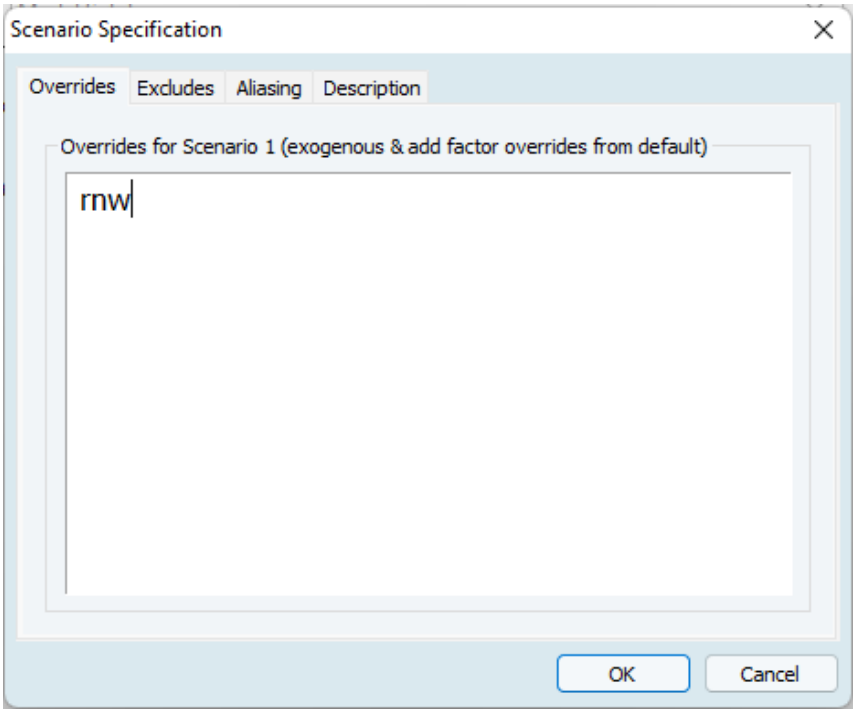


Now, for the alternate scenario (Scenario 1), click on "Edit Scenario Options" to obtain:

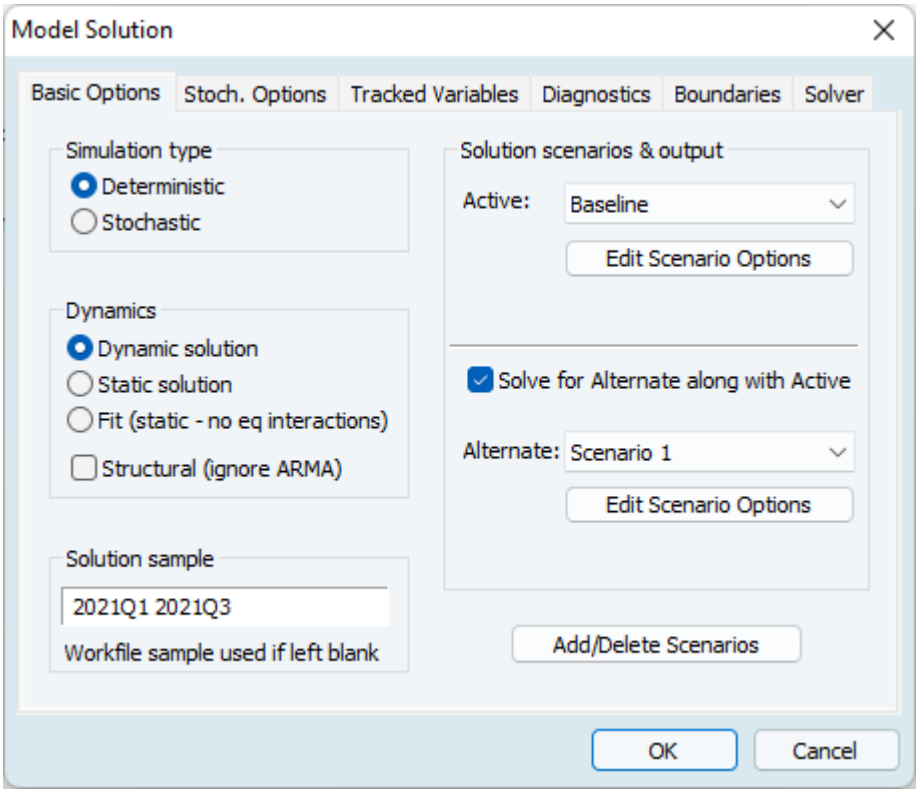


Note that, by convention, when computing the alternative scenario EViews expects to find the alternative scenario in a variable with "_#" (where # is replaced with the scenario number). However, it will use this variable in the alternate scenario only if you instruct it to do so explicitly using the **"Scenario Specification"** dialog box.

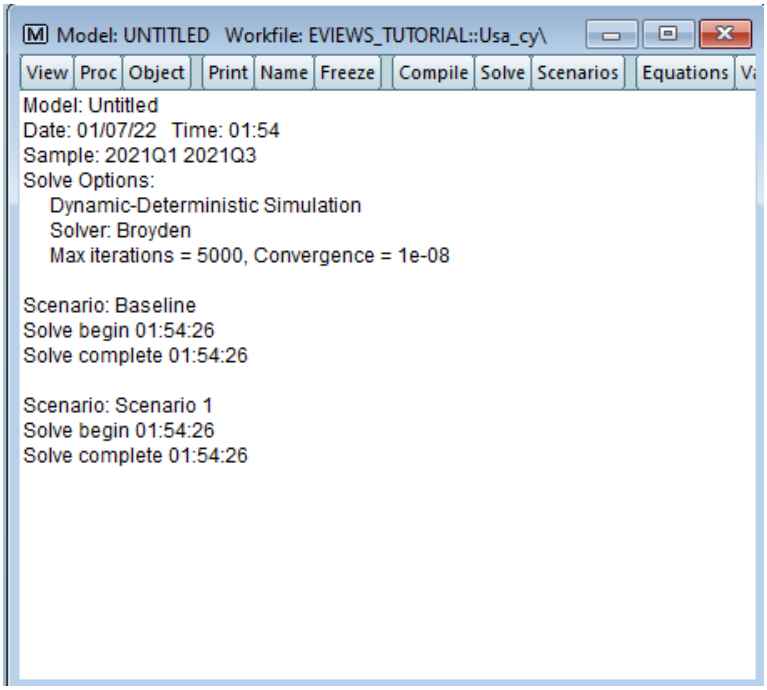
The "Overrides for Scenario 1" text box expects the names of the variables that will be changed during the computation of the alternative scenario. In this case, you need to enter "**rnw**", which, by EViews's convention, will be overwritten by "**rnw_1**" during the computation of the alternative scenario. Recall that **rnw_1** is 10 percent higher than **rnw** during 2021Q1-2021Q3.



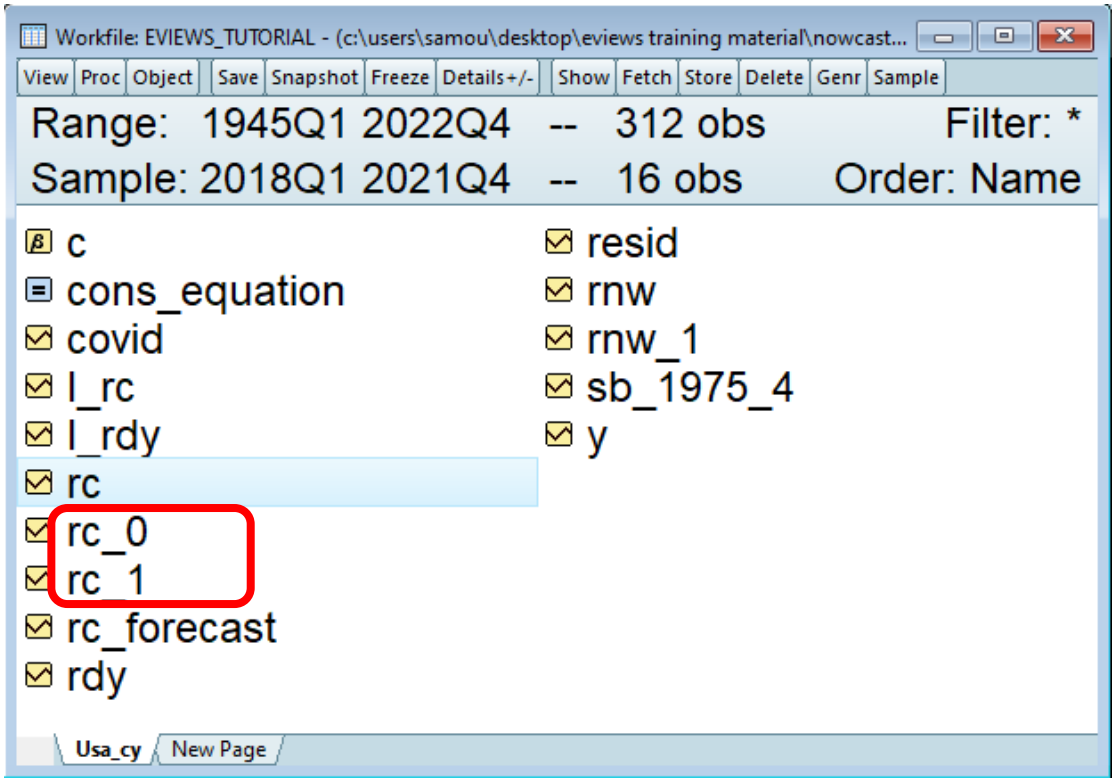
Now press OK to return to the solve menu:



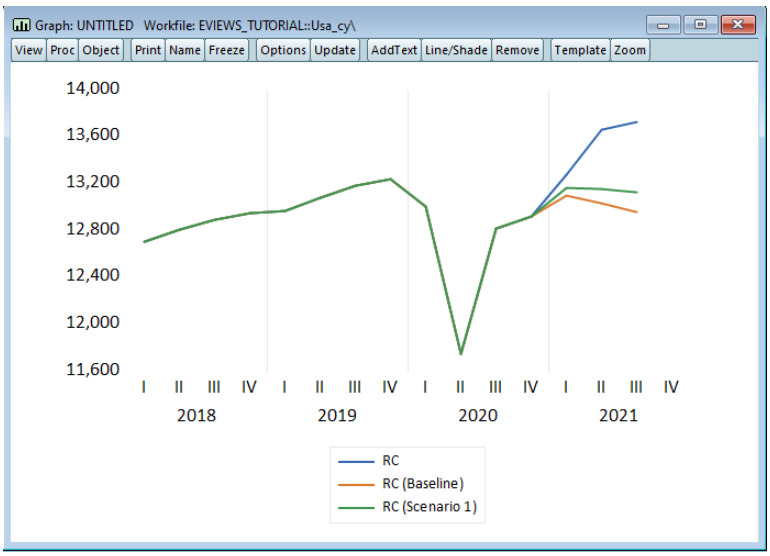
Now press OK to solve the model for the baseline forecast and the alternative forecast that assumes volume will be 10 percent higher during 2021Q1-2021Q3. You should see the following output, which confirms that there were no numerical errors during the calculation of both the baseline and scenario 1.



The model simulator creates (or overwrites) two variables in the current pagefile, **usa_cy**: **rc_0** (baseline) and **rc_1** (value of real consumption under the alternate scenario).

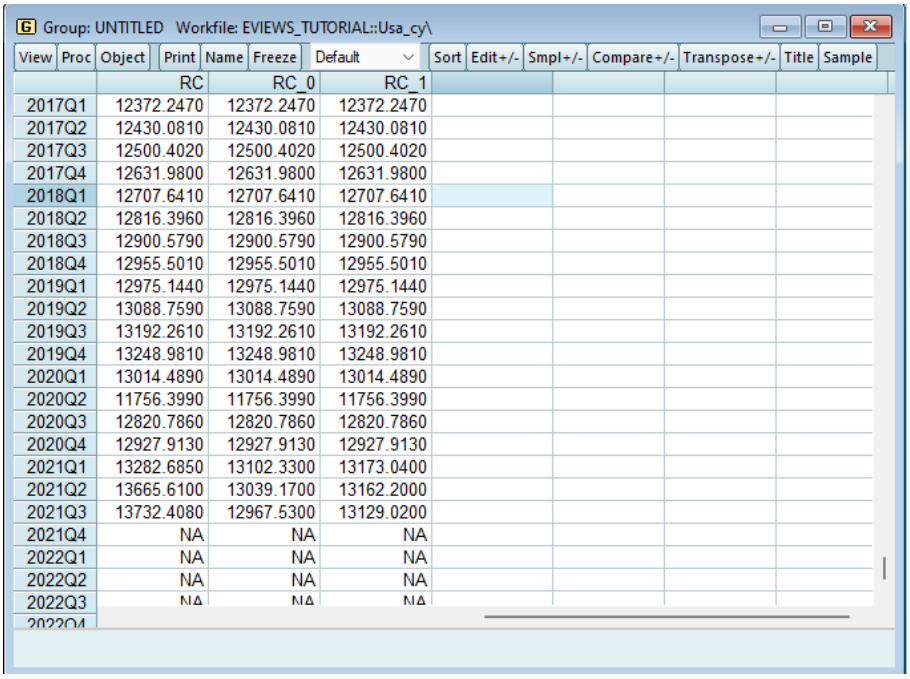


To see the impact on the alternative assumption on the forecast, you can issue: **plot rc rc_0 rc_1**



The positive impact of the 10 percent shock to real net worth is quite evident for 2021Q1-2021Q3.

Alternatively, you may choose to simply view the corresponding data: **show rc rc_0 rc_1**



Group: UNTITLED Workfile: EVIEWS_TUTORIAL::Usa_cy\

View | Proc | Object | Print | Name | Freeze | Default | Sort | Edit+/- | Smpl+/- | Compare+/- | Transpose+/- | Title | Sample

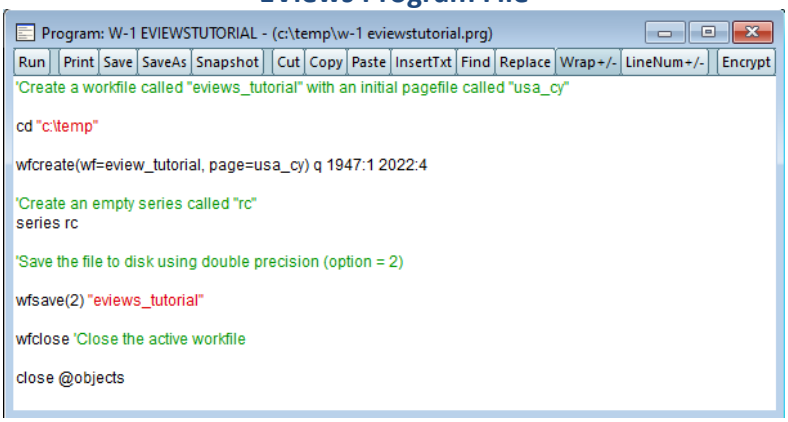
	RC	RC_0	RC_1
2017Q1	12372.2470	12372.2470	12372.2470
2017Q2	12430.0810	12430.0810	12430.0810
2017Q3	12500.4020	12500.4020	12500.4020
2017Q4	12631.9800	12631.9800	12631.9800
2018Q1	12707.6410	12707.6410	12707.6410
2018Q2	12816.3960	12816.3960	12816.3960
2018Q3	12900.5790	12900.5790	12900.5790
2018Q4	12955.5010	12955.5010	12955.5010
2019Q1	12975.1440	12975.1440	12975.1440
2019Q2	13088.7590	13088.7590	13088.7590
2019Q3	13192.2610	13192.2610	13192.2610
2019Q4	13248.9810	13248.9810	13248.9810
2020Q1	13014.4890	13014.4890	13014.4890
2020Q2	11756.3990	11756.3990	11756.3990
2020Q3	12820.7860	12820.7860	12820.7860
2020Q4	12927.9130	12927.9130	12927.9130
2021Q1	13282.6850	13102.3300	13173.0400
2021Q2	13665.6100	13039.1700	13162.2000
2021Q3	13732.4080	12967.5300	13129.0200
2021Q4	NA	NA	NA
2022Q1	NA	NA	NA
2022Q2	NA	NA	NA
2022Q3	NA	NA	NA
2022Q4			

The model simulator might seem overkill to produce a forecast for a single equation. However, the model simulator can handle large scale (multiple equation) models with far richer, and multiple alternative scenarios. The model solver offers the most general approach to producing forecasts using EViews.

Section 6: EViews Program Files

A program file is a sequence of EViews commands that will be executed consecutively.[7] You can create a program file by clicking **"File" => "New" => "Program".** You may use one single quotation mark to add comments or notes to your command line. To execute the program file, click "**Run**".

EViews Program File



Program: W-1 EVIEWSTUTORIAL - (c:\temp\w-1 eviewstutorial.prg)

Run | Print | Save | SaveAs | Snapshot | Cut | Copy | Paste | InsertTxt | Find | Replace | Wrap+/- | LineNum+/- | Encrypt

```
'Create a workfile called "eviews_tutorial" with an initial pagefile called "usa_cy"

cd "c:\temp"

wfcreate(wf=eview_tutorial, page=usa_cy) q 1947:1 2022:4

'Create an empty series called "rc"
series rc

'Save the file to disk using double precision (option = 2)

wfsave(2) "eviews_tutorial"

wfclose 'Close the active workfile

close @objects
```

Open the file called **eviews_tutorial.prg** and click on the "Run" button to reproduce all the results in this tutorial.

[7] A very succinct guide on the EViews programming can be found at Programming (eviews.com)

1. PROGRAM BASICS

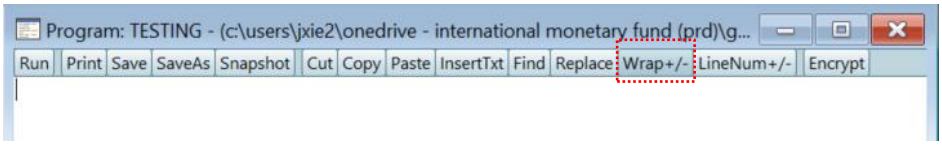
EViews' programming features allow you to create or store commands in programs that automate repetitive tasks and generate a record of the steps necessary to replicate your research project.

1) **Creating a program:** there are two ways to create a program file in EViews.

   a. Using the user interface: click **File/New/Program**.
   b. Using the command window: type "program", followed by an optional program name. For example**,** the command **"**program testing" opens a program window named "TESTING"

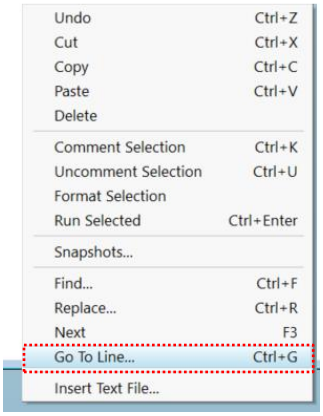
2) **Program Formatting**: an EViews program is a standard ASCII text file, consisting of one or more lines of text. Each line generally represents a single EViews command. A few rules/tips to note for the formatting:

   a. **Autowrap**: if a program line is longer than the current program window, EViews will automatically wrap the text of the line, but does not change its content. If the autowrap doesn't appear automatically, click Wrap +/- on the menu bar to turn on the function.

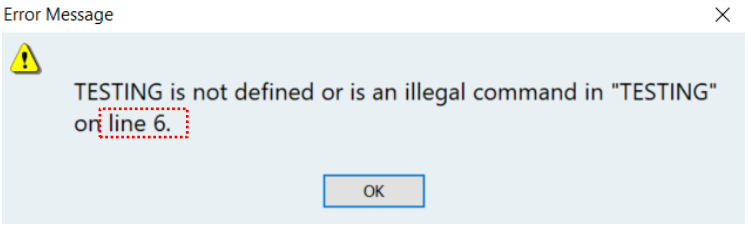


   b. **Display line numbers**: when autowrapping is turned off via the option menu, one may elect to show program line numbers in the program window by clicking LineNum +/-.

      i. Right clicking anywhere in your program and selecting Go To Line… will jump directly to a specific line number



      ii. This function is used to locate the specific line where an error occurs. An error example is shown below.



c. **Break long lines manually**: use the underscore character "_" as the last character on the line to join multiple lines. For example, the three separate lines of text

```
equation eq1.ls _
y x c _
ar(1) ar(2)
```

is equivalent to the single line

```
equation eq1.ls y x c ar(1) ar(2)
```

d. The apostrophe " ' " is the **comment character** in the program which treats all characters that follow as a comment. The comment line will be ignored when executing the program command.

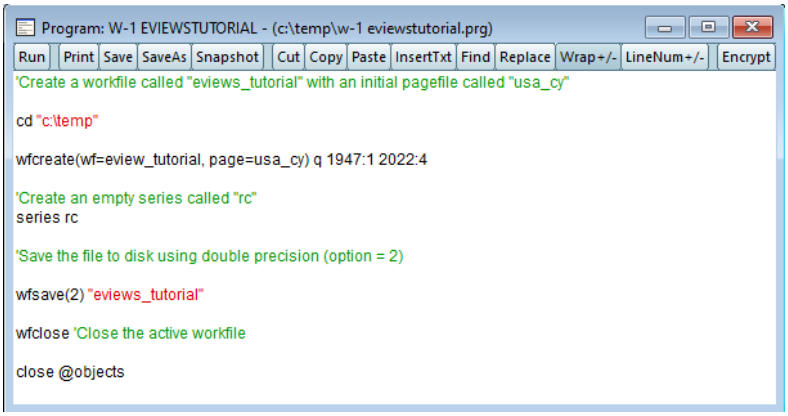
3) **Executing a program**: A program file is a sequence of EViews commands that will be executed consecutively.[8]

a. To execute the program file, you may click **Run**.

b. Running this program creates a new workfile called **EViews_Tutorial.wf1**. It has a single pagefile called **usa_cy**, and a series **rc** (with missing values for the entire workfile range).

c. Please open the file called **eviews_tutorial.prg**. Click on the Run button to reproduce all the results in this tutorial.

[8] A very succinct guide on the EViews programming can be found at Programming (eviews.com)



4) **Stopping a Program**: the following commands or keys will stop program execution.

   a. Pressing **ESC** or the **F1** key**:** halts the execution of a program. It may take a few seconds for EViews to respond to the halt command.

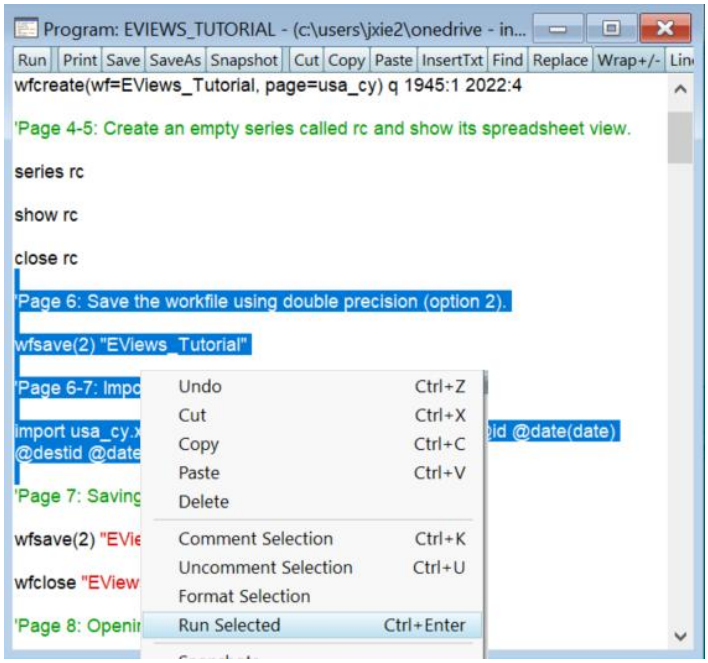
   b. Using the **Stop** command: programs will stop when they encounter this command.

   c. Using the **Exit** keyword (rarely used): the EViews application will close.

5) **Running part of a program**: you may choose to run only part of your program by highlighting certain lines, then right-clicking and selecting Run Selected. There are, however, some potential pitfalls when using Run Selected:

   a. No arguments or options can be passed into the selected lines of code. Meaning that if the selected lines rely on program arguments or options, they will fail to execute properly.

   b. Any program variables declared outside of the selected lines may not be initialized properly when the selected lines are run.



2. SIMPLE PROGRAMS

The command language in EViews can be split into four different groups:

1) **General commands**: create, structure, and manipulate workfiles (including importing and exporting data) and workfile objects.

2) **Object commands:** execute views and procedures on objects in the workfile (such as series, groups, and equations).

3) **Functions commands**: create new data, either in the form of workfile objects or as program variables. Such commands usually start with a "@" symbol. For example, @mean(rc) will return the average value of the series rc.

4) **Object data members commands**: retrieve a fundamental piece of data from an object. This command provides access to existing results and usually starts with an "@" symbol. It can be accessed by typing the name of the object followed by a "." and the name of the data member. For example, eq1.@coefs returns a vector containing the estimated coefficients from equation eq1.

3. PROGRAM VARIABLES

While one can use programs to run and re-run a collection of EViews commands, the real power of the EViews programming language comes from the use of *program variables* and *program control statements*.

Program variables are variables in place of numeric or string values in the EViews programs. There are two basic types of program variables: *control (numeric) variables* and *string variables*.

1) **Control variable:** used in place of numeric values in an EViews program.

    a. A control variable starts with an "!" character followed by its assigned name. You may assign a value to the control variable using a "=" sign and a numerical number. For example:

        !x = 8

        Once assigned the value, a control variable may appear in an expression, for example:

        genr y = !x/2

    b. Once a control variable is assigned a value, one can place it in a program wherever a number would appear.
    c. Control variables do NOT exist outside of your program and are automatically deleted after the program finishes.
    d. An alternative to a control variable is to create a scalar object in the EViews workfile.

2) **String variable:** use in place of string expression in an EViews program.

    a. A string variable is a program variable whose value is a string of text. String variables have names that begin with a "%" symbol and can be assigned to a string expression using "=" sign. For example:

        %regressors = "credit_card ip pmi"

    b. String operations can concatenate the contents of multiple string and control variables. For example:

        !rgdp_growth = 5
        %st1 = "Real GDP growth rate"
        %st2 = "Namibia"
        %st3 = %st1 + " of " + %st2 + " is " + @str(!rgdp_growth) + " %"

        The above lines will generate a string variable "Real GDP growth rate of Namibia is 5 %"

3) **Replacement Variables**: use replacement variables as an indirect way of referring to something (a command, an object name, or a portion of names for one or more items).

    a. Enclosing a string variable in curly braces ("{" and "}"), will replace the expression with an object whose name is given by the string value. For example:

        %x = "gdp"

```
series y = {%x}
```

These program lines will generate a new series “y” using the time series object “gdp”.

b. Replacement variables may be constructed using nested evaluation of replacement variables. For example:

```
%str1 = “x”
%name = “%str1”
```

Then one can declare the series X and fill it with random normal values using the command:

```
series {{%name}} = nrnd
```

The double braces will result in a double replacement, yielding:

```
series x = nrmc
```

4. CONTROL OF EXECUTION

EViews provides several ways to control the execution of commands in a program. Controlling execution means executing commands selectively or repeatedly under specific conditions.

1) “IF” statements: only execute commands if some condition is satisfied.

a. An IF statement starts with the IF keyword, followed by an expression for the condition, and the word THEN. Use ENDIF to close the IF statement. For example:

```
if !scale > 0 then
   series newage = age/!scale
endif

‘Execute the series statement if !SCALE is larger than 0.
```

b. One may add AND/OR statements in the condition, using parentheses to group parts of the statement if necessary. For example:

c. One may also apply multiple IF statements after ELSE, and to string variables. Note the string comparisons are case sensitive. For example:

```
If ctry = “CA” or ctry = “IN” then
     series ctryid = 1
else
     If ctry = “CN” then
```

```
            series ctryid = 2
        Endif
    endif
```

d. There are other relationship operators: =, >, >=, <, <=, <>

2) The "FOR" loop: repeat a set of commands for different values of numerical or string variables.

a. A FOR loop begins with a FOR statement and ends with a NEXT statement. The syntax of the FOR statement differs depending upon whether it uses numerical variables or string variables.

b. ***FOR Loops with Numerical Control Variables or Scalars***:

i. To repeat statements for different values of a control variable, use the FOR keyword with a control variable initialization, the word TO, and then an ending value. For example:

```
for !j = 1 to 10
    series y_{!j} = rgdp_{!j}
next
```

ii. After the ending value, adding the word STEP followed by a number allows varying an amount of change in the control variable each time the loop is executed. Without including STEP, the default is 1. For example:

```
for !j = 1 to 10 step 1
    series y_{!j} = rgdp_{!j}
next
```

c. ***FOR loops with string variables and string objects:***

i. To repeat statements for different values of a string variable, use the FOR loop to move over a list of string values. Use FOR keyword, followed by the name of the string program variable, followed by the list of values. For example:

```
for %y gdp gnp ndp nnp
    equation {%y}.ls {%y} c {%y}(-1)
next
```

ii. Both string objects and replacement variables may be used to specify a list for use in loops, by surrounding the object name or replacement variables with curly braces ("{ }"). For example:

```
string variables = "gdp gnp ndp nnp"
```

```
for %y {variables}
    equation {%y}.ls {%y} c {%y}(-1)
next
```

3) **The "WHILE" loop**: allows us to repeat a series of commands several times, but only while one or more conditions are satisfied.

   a. The while loop begins with a WHILE statement and ends with a WEND statement. For example:

```
!var = 1
!a = 1
while !val < 10000 and !a < 10
    series inc{!val} = income/!val
    !val = !val * 10
    !a = !a + 1
wend
```

   b. Unlike a FOR statement, the WHILE statement does not update the control variable used in the test condition. One needs to include an explicit statement inside the loop to change the control variable, or else the loop will never terminate (Note: you can use the F1 or ESC key to break out of an infinite loop).